

$$2 \left[\cos 1 - \frac{14}{15} + \frac{4}{15} \right]$$

$$2 \left[\cos 1 - \frac{2}{3} \right] = -ve \text{ number}$$

$$h(x) = f(x) + x^2 g(x)$$

$$h'(x) = \sqrt{2} \cos \left(x + \frac{\pi}{4} \right) + \frac{8x^3}{3}$$

$$= \sin x + \cos x - \frac{x^4}{3} + x^2 + x^4 - x^2$$

$$= \sqrt{2} \sin \left(x + \frac{\pi}{4} \right) + \frac{2x^4}{3}$$

$\Rightarrow h(x)$ is decreasing and then it is always increasing

$\therefore h(x)$ has minima but not maxima.

Paragraph for Question I02/3/099 to I02/3/100

$$\text{Let } I_1 = \int_0^x \frac{t^2 dt}{(x - \sin x) \sqrt{\lambda^2 + 2\lambda t}} \text{ (where } \lambda \text{ is a non-zero constant) and } I_2 = \int_0^{x^2} \frac{e^t - 1 - t}{x^6} dt.$$

On the basis of above information, answer the following questions :

$$\text{माना } I_1 = \int_0^x \frac{t^2 dt}{(x - \sin x) \sqrt{\lambda^2 + 2\lambda t}} \text{ (जहाँ } \lambda \text{ अशून्य अचर है) तथा } I_2 = \int_0^{x^2} \frac{e^t - 1 - t}{x^6} dt \text{ है।}$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए। :

I02/3/099. If $\lim_{x \rightarrow 0} I_1 = 1$, then the sum of value(s) of λ is equal to :

यदि $\lim_{x \rightarrow 0} I_1 = 1$ हो, तो λ के मानों का योगफल होगा :

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

Ans. A

Sol.

I02/3/100. If $\lambda = -1$, then the value of $\lim_{x \rightarrow 0} (I_1 + 6I_2)$ is equal to :

यदि $\lambda = -1$ हो, तो $\lim_{x \rightarrow 0} (I_1 + 6I_2)$ का मान होगा :

- (A) 2 (B) 3 (C) -1 (D) 0

Ans. B

Sol.
$$\lim_{x \rightarrow 0} I_1 = \frac{\lim_{x \rightarrow 0} \int_0^x \frac{t^2 dt}{\sqrt{\lambda^2 + 2\lambda t}}}{(x - \sin x)} = \lim_{x \rightarrow 0} \frac{x^2}{(1 - \cos x)\sqrt{\lambda^2 + 2\lambda x}} = 1$$

$$\Rightarrow \frac{2}{|\lambda|} = 1 \Rightarrow |\lambda| = 2 \quad \lambda = \pm 2$$

$\therefore$ sum of values of $\lambda = 0$.

$$\begin{aligned} \lim_{x \rightarrow 0} I_2 &= \lim_{x \rightarrow 0} \int_0^{x^2} \frac{e^t - 1 - t}{x^6} dt = \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 - x^2}{6x^5} 2x \\ &= \frac{1}{3} \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 - x^2}{(x^2)^2} = \frac{1}{6} \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} (I_1 + 6I_2) = \frac{2}{1} + 1 = 3$$

Paragraph for Question I02/3/101 to I02/3/102

Let f be a cubic polynomial function such that $f(x) + f(-x) = 0 \forall x \in \mathbb{R}$ and $\lim_{x \rightarrow 0} \left(\frac{f(x)}{x} + 1 \right)^{\frac{1}{x^2}} = e^4$.

On the basis of above information, answer the following questions :

माना f बहुपदीय घन फलन इस प्रकार है कि $f(x) + f(-x) = 0 \forall x \in \mathbb{R}$ तथा $\lim_{x \rightarrow 0} \left(\frac{f(x)}{x} + 1 \right)^{\frac{1}{x^2}} = e^4$ है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I02/3/101. The value of $\sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2r}{f(r)} \right)$ equals :

$\sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2r}{f(r)} \right)$ का मान होगा :

- (A) $\cot^{-1} 1 + \cot^{-1} 2$ (B) $\tan^{-1} 1 + \tan^{-1} 2$ (C) $\tan^{-1} 2 + \tan^{-1} 3$ (D) $\cot^{-1} 2 + \cot^{-1} 3$

Ans. D

Sol.

I02/3/102. If $g(f(x)) = x \forall x \in \mathbb{R}$ and $g(4) = m$ and $g(32) = n$, then $\int_m^n f(x)dx + \int_4^{32} g(x)dx$ is equal to :

यदि $g(f(x)) = x \forall x \in \mathbb{R}$, $g(4) = m$ तथा $g(32) = n$ हो, तो $\int_m^n f(x)dx + \int_4^{32} g(x)dx$ का मान होगा :

- (A) 28 (B) 60 (C) 30 (D) 16

Ans. B

Sol.

$$\text{Given } \lim_{x \rightarrow 0} \left(\frac{f(x)}{x} + 1 \right)^{\frac{x^2}{x}} = e^4$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} \frac{f(x)}{x^3}} = 4 \Rightarrow f(x) = 4x^3$$

$$\therefore \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2r}{f(r)} \right) = \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2r}{4r^3} \right) = \sum_{r=1}^{\infty} \tan^{-1} \frac{2}{1 + (4r^2 - 1)}$$

$$= \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{(2r+1) - (2r-1)}{1 + (2r+1)(2r-1)} \right)$$

$$= \sum_{r=1}^{\infty} (\tan^{-1}(2r+1) - \tan^{-1}(2r-1))$$

$$= \lim_{n \rightarrow \infty} \left[(\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 5 - \tan^{-1} 3) + (\tan^{-1} 7 - \tan^{-1} 5) + \dots + (\tan^{-1}(2n+1) - \tan^{-1}(2n-1)) \right]$$

$$= \lim_{n \rightarrow \infty} [\tan^{-1}(2n+1) - \tan^{-1} 1]$$

$$= \frac{\pi}{2} - \tan^{-1} 1 = \frac{\pi}{4}$$

$$\text{but } \cot^{-1} 2 + \cot^{-1} 3 = \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$$

$$= \tan^{-1} \left\{ \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} \right\} = \tan^{-1} 1 = \frac{\pi}{4}$$

Paragraph for Question I02/3/103 to I02/3/105

Consider $f(x) = \sin x + \int_0^x (x-t)f(t)dt$.

On the basis of above information, answer the following questions

माना $f(x) = \sin x + \int_0^x (x-t)f(t)dt$ है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए।

I02/3/103. The value of $\int_0^\pi (f''(x) - f(x))dx$ is equal to :

$\int_0^\pi (f''(x) - f(x))dx$ का मान होगा :

- (A) 2 (B) 1 (C) -1 (D) -2

Ans. D

Sol.

I02/3/104. The period of the function $g(x) = f(x) - f''(x)$ is :

फलन $g(x) = f(x) - f''(x)$ का आवर्तकाल होगा :

- (A) $\frac{\pi}{4}$ (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

Ans. B

Sol.

I02/3/105. The value of $\int_{\pi/2}^{3\pi/2} \sin^{-1}(f(x) - f''(x))dx$ is equal to :

$\int_{\pi/2}^{3\pi/2} \sin^{-1}(f(x) - f''(x)) dx$ का मान होगा :

- (A) π (B) 2π (C) 0 (D) 3π

Ans. C

Sol. Consider $f(x) = \sin x + \int_0^x (x-t)f(t)dt$

$$f(x) = \sin x + x \int_0^x f(t)dt - \int_0^x tf(t)dt$$

$$f'(x) = \cos x + xf(x) + \int_0^x f(t)dt - xf(x)$$

$$f''(x) = -\sin x + f(x)$$

$$f''(x) - f(x) = -\sin x$$

$$\sin^{-1}(f(x) - f''(x)) = \sin^{-1}(\sin x)$$

$$\int_{\pi/2}^{3\pi/2} \sin^{-1}(\sin x) dx = \int_{\pi/2}^{3\pi/2} (\pi - x) dx$$

$$= \pi^2 - \frac{1}{2} \left(\frac{9\pi^2}{4} - \frac{\pi^2}{4} \right)$$

$$\pi^2 - \pi^2 = 0$$

Paragraph for Question I02/3/106 to I02/3/108

$$\text{Let } f(x) = \frac{x^2 - 7x + 6}{x^2 - 4x + b}$$

On the basis of above information, answer the following questions :

$$\text{माना } f(x) = \frac{x^2 - 7x + 6}{x^2 - 4x + b} \text{ है।}$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए।

I02/3/106. If $y = f(x)$ has two asymptotes, then sum of integral values of b is :

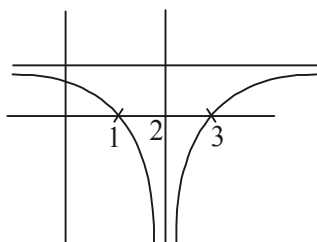
यदि $y = f(x)$ की दो अनन्तस्पर्शियाँ हो, तो b के पूर्णांक मानों का योगफल होगा :

- (A) -5 (B) 4 (C) 7 (D) 10

Ans. A

Sol.

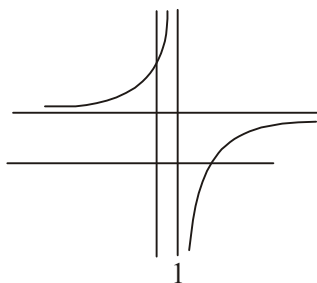
$$f(x) = \frac{x^2 - 7x + 6}{x^2 - 4x + 6}$$



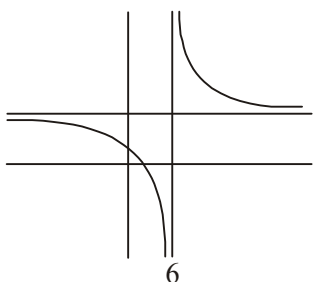
$$b = 4$$

Common root

$$y = \frac{(x-6)(x-1)}{(x-1)(x-3)} \quad b = 3$$



$$y = \frac{(x-6)(x-1)}{(x-6)(x+2)} \quad b = -12$$



sum of values of $b = -5$

102/3/107. If $b = 3$, then value of $\int_4^7 f(x) dx$ is :

यदि $b = 3$ हो, तो $\int_4^7 f(x) dx$ का मान होगा :

(A) $3(1 - \ln 3)$

(B) 0

(C) $3(1 - \ln 7)$

(D) $3(1 - \ln 4)$

Sol.

$$\begin{aligned}
 \int_4^7 f(x) dx &= \int_4^7 \frac{x^2 - 7x + 6}{x^2 - 4x + 3} dx \\
 &= \int_4^7 \frac{(x-6)(x-1)}{(x-3)(x-1)} dx \\
 &= \int_4^7 \left(\frac{x-3-3}{x-3} \right) dx \\
 &= \int_4^7 \left[1 - \frac{3}{x-3} \right] dx \\
 &= 3 - 3 \left[\ell n^{x-3} \right]_4^7 \\
 &= 3 - 3 \left[\ell n^4 - \ell n^1 \right] \\
 &= 3 - 3 \ell n^4 \\
 &= 3(1 - \ell n^4)
 \end{aligned}$$

I02/3/108. If $b = 4$ and $f(x) = k$ has 2 solutions, then number of positive integral values of (k) is :

यदि $b = 4$ तथा $f(x) = k$ के 2 हल हो, तो (k) के धनात्मक पूर्णांक मानों की संख्या होगी :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol.

$$f(x) = k = \frac{x^2 - 7x + 6}{x^2 - 4x + 4}$$

$$\Rightarrow x^2 - 7x + 6 = k(x^2 - 4x + 4)$$

$$(k-1)x^2 + (-4k+7)x + (4k-6) = 0$$

For two solutions

$$k-1 \neq 0 \text{ and } (7-4k)^2 - 4(4k-6) = 0$$

$$(49 + 16k^2 - 56k - 16k^2 + 40k - 24) > 0$$

$$-16k + 25 > 0$$

$$\left(k < \frac{25}{16} \right)$$

$$\Rightarrow k = 1$$

put when $k = 1$, we don't have two solutions.

$$\Rightarrow \text{Number of positive integral value of } k = 0$$

Paragraph for Question I02/3/109 to I02/3/111

Let f, g and h be three real-valued differentiable functions on $\mathbb{R}^+$ satisfying $\int_0^x f(t) dt = x + \ell n x - \int_x^0 \sin^2 t f(t) dt$,

$$g(x) = \lim_{x \rightarrow 0} \left(\lim_{a \rightarrow 0} \frac{1}{(\sin a)^4} \int_0^a \frac{(e^{e^{-f(x)} + \sin b} - e^{e^{-f(x)}}) \ell n^2 (1 + \sin b) db}{\left(\sin^5 b + \frac{1}{4} \right)} \right) \& h(x) = \cos^2 x.$$

On the basis of above information, answer the following questions :

माना f, g तथा $h, \mathbb{R}^+$ पर तीन वास्तविक अवकलनीय फलन है जो इस प्रकार है कि $\int_0^x f(t) dt = x + \ell n x - \int_x^0 \sin^2 t f(t) dt$,

$$g(x) = \lim_{x \rightarrow 0} \left(\lim_{a \rightarrow 0} \frac{1}{(\sin a)^4} \int_0^a \frac{(e^{e^{-f(x)} + \sin b} - e^{e^{-f(x)}}) \ell n^2 (1 + \sin b) db}{\left(\sin^5 b + \frac{1}{4} \right)} \right) \text{ और } h(x) = \cos^2 x.$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I02/3/109. The value of $\int_0^{\pi/4} x f(x) dx$ is less than :

$\int_0^{\pi/4} x f(x) dx$ का मान से कम है :

- (A) -1 (B) 0 (C) 1 (D) 2

Ans. D

Sol. $f(x) = 1 + \frac{1}{x} + \sin^2(x) f(x)$

$$\Rightarrow f(x) = \frac{\sec^2 x}{x} + \sec^2 x$$

Domain of $f(x) = \mathbb{R}^+ - \{(2n+1)\pi/2\}$

$$g(x) = \lim_{x \rightarrow 0^+} e^{-n(x)} \left(\lim_{a \rightarrow 0} \frac{1}{(\sin a)^4} \int_0^a \frac{(e^{\sin b} - 1) / n^2 (1 + \sin b)}{\left(\sin^5 b + \frac{1}{4} \right)} \right)$$

$$f(0^+) \rightarrow \infty$$

$$g(x) = \lim_{a \rightarrow 0} \frac{(e^{\sin a} - 1) \ell n^2 (1 + \sin a)}{4(\sin a)^3 \left(\sin^5 a + \frac{1}{4} \right)} = 1$$

$$\int_0^{\pi/4} x \left(\frac{\sec^2 x}{x} + \sec^2 x \right) dx$$

$$= \int_0^{\pi/4} (\sec^2 x + x \sec^2 x) dx$$

$$= [(1+x) \tan x - \ln(\sec x)]_0^{\pi/4}$$

$$= 1 + \frac{\pi}{4} - \frac{1}{2} \ln 2 < 2$$

I02/3/110. The number of solutions of the equation $\sqrt{h(x)} \cdot f(g(x)) = 1$ in the interval $x \in (0, 2\pi]$ is :

समीकरण $\sqrt{h(x)} \cdot f(g(x)) = 1$ के अन्तराल $x \in (0, 2\pi]$ में हलों की संख्या है

(A) 0

(B) 2

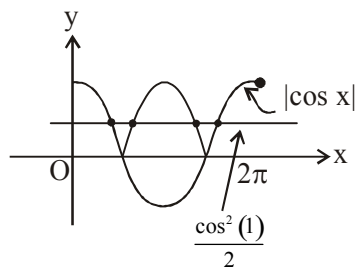
(C) 3

(D) 4

Ans.

D

Sol.



$$|\cos x| \cdot 2 \sec_2(1) = 1$$

$$\Rightarrow |\cos x| = \frac{\cos^2(1)}{2}$$

I02/3/111. Choose the wrong option :

- (A) $f(x)g(x)h(x)$ is strictly decreasing function in its domain
 (B) $f(x)g(x)h(x)$ does not attain local minima.
 (C) $f(x)g(x)h(x)$ is neither odd nor even function.
 (D) The domain of $f(x)g(x)h(x)$ is $\mathbb{R}^+$

गलत कथन चुनो :

- (A) $f(x)g(x)h(x)$ अपने प्रांत में निरन्तर ह्रस्वमान फलन है
 (B) $f(x)g(x)h(x)$ कोई स्थानीय निम्निष्ठ मान नहीं रखते है
 (C) $f(x)g(x)h(x)$ ना तो विषम है और ना ही सम है
 (D) फलन $f(x)g(x)h(x)$ का प्रांत $\mathbb{R}^+$ है

Ans. D

Sol. Domain of $f(x)g(x)h(x)$

$$= \left(\frac{\sec^2 x}{x} + \sec^2 x \right) \cdot 1 \cdot \cos^2 x$$

$$\text{is } \mathbb{R}^+ - \left\{ (2n+1) \frac{\pi}{2} \right\}$$

Paragraph for Question I02/3/112 to I02/3/113

Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(a) = b$ and $f'(a) = f''(a) = 0$. Suppose $g(x) = f(x) - f(a) + (a-x)f'(x) + 3(x-a)^2$ for which conclusion of Rolle's theorem in $[a, b]$ holds at $x = 2$, where $2 \in (a, b)$.

माना कि $f(x)$ एक तीन घात का बहुपद है जिसका अग्रणी गुणांक इकाई है। इस प्रकार से है कि $f(a) = b$ तथा $f'(a) = f''(a) = 0$ हैं। माना कि $g(x) = f(x) - f(a) + (a-x)f'(x) + 3(x-a)^2$ जिसके लिए रोले प्रमेय को अंतराल $[a, b]$ में $x = 2$ पर संतुष्ट करता है जहाँ $2 \in (a, b)$.

I02/3/112. The value of $f''(2)$ is equal to :

$f''(2)$ का मान बराबर है :

- (A) 2 (B) 3 (C) 4 (D) 6

Ans. D

Sol. Let $f(x) = x^3 + Px^2 + Qx + R$

$$f''(a) = 6a + 2P = 0 \Rightarrow P = -3a$$

$$f'(a) = 3a^2 + 2Pa + Q = 0 \Rightarrow Q = 3a^2$$

$$f(a) = b \Rightarrow b = R + a^3 \Rightarrow R = b - a^3$$

$$\Rightarrow f(x) = (x-a)^3 + b$$

$$\Rightarrow g(x) = (x-a)^3 - 3(x-a)^3 + 3(x-a)^2 = (x-a)^2 (2a+3-2x)$$

$$\Rightarrow g(x) = -2(x-a)^2 \left(x - \left(a + \frac{3}{2} \right) \right)$$

Rolle's theorem holds for g in [a, b] at x = 2

$$\Rightarrow g(a) = g(b) \text{ and } g'(2) = 0 \text{ with } 2 \in (a, b)$$

$$\Rightarrow g(b) = 0 \Rightarrow b = a + \frac{3}{2}$$

$$g'(x) = -2(x-a) \left[2 \left(x - \left(a + \frac{3}{2} \right) \right) + (x-a) \right]$$

$$= -2(x-a) [2x - (2a+3) + x-a]$$

$$= -2(x-a) [3x - (3a+3)] = -6(x-a)(x-(a+1))$$

$$\Rightarrow a+1 = 2 \text{ and } a = 1$$

$$f''(x) = 6(x-a) \Rightarrow f''(2) = 6$$

I02/3/113. The value of definite integral $\int_a^b f(x) dx$ is equal to :

निश्चित समाकलन $\int_a^b f(x) dx$ का मान बराबर होगा :

(A) $\frac{123}{64}$

(B) $\frac{213}{64}$

(C) $\frac{321}{64}$

(D) $\frac{481}{64}$

Ans. C

Sol. Let $f(x) = x^3 + Px^2 + Qx + R$

$$f''(a) = 6a + 2P = 0 \Rightarrow P = -3a$$

$$f'(a) = 3a^2 + 2Pa + Q = 0 \Rightarrow Q = 3a^2$$

$$f(a) = b \Rightarrow b = R + a^3 \Rightarrow R = b - a^3$$

$$\Rightarrow f(x) = (x-a)^3 + b$$

$$\Rightarrow g(x) = (x-a)^3 - 3(x-a)^3 + 3(x-a)^2 = (x-a)^2 (2a+3-2x)$$

$$\Rightarrow g(x) = -2(x-a)^2 \left(x - \left(a + \frac{3}{2} \right) \right)$$

Rolle's theorem holds for g in [a, b] at x = 2

$$\Rightarrow g(a) = g(b) \text{ and } g'(2) = 0 \text{ with } 2 \in (a, b)$$

$$\Rightarrow g(b) = 0 \Rightarrow b = a + \frac{3}{2}$$

$$\begin{aligned}
 g'(x) &= -2(x-a) \left[2 \left(x - \left(a + \frac{3}{2} \right) \right) + (x-a) \right] \\
 &= -2(x-a) [2x - (2a+3) + x-a] \\
 &= -2(x-a) [3x - (3a+3)] = -6(x-a)(x-(a+1)) \\
 &\Rightarrow a+1=2 \text{ and } a=1
 \end{aligned}$$

$$\int_a^b f(x) dx = \int_a^b (x-a)^3 + b dx$$

$$= \int_0^{b-a} x^3 + b dx = \int_0^{\frac{3}{2}} x^3 + \frac{5}{2} dx = \frac{81}{64} + \frac{15}{4} = \frac{321}{64}$$

Paragraph for Question I03/3/114 to I03/3/115

A conic C passing through P(1, 2) is such that the slope of its tangent at any point on the conic is inversely proportional to the ordinate of that point and conic C passes through origin.

On the basis of above information, answer the following questions :

एक शांकव C बिन्दु P(1,2) से इस प्रकार गुजरता है कि शांकव के किसी बिन्दु पर स्पर्श रेखा की प्रवणता उस बिन्दु की कोटि के व्युत्क्रमानुपाती है तथा शांकव C मूल बिन्दु से गुजरता है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I03/3/114. If 'm' be the slope of the common tangent of the conic C and the circle $x^2 + y^2 = 1$ then m^2 is :

यदि शांकव C तथा वृत्त $x^2 + y^2 = 1$ की उभयनिष्ठ स्पर्श रेखा की प्रवणता m हो, तो m^2 का मान होगा :

(A) $2 \sin \frac{\pi}{10}$ (B) $2 \cos \frac{\pi}{5}$ (C) $2 \tan \frac{\pi}{8}$ (D) $2 \cot \frac{\pi}{8}$

Ans. A

Sol. Given $\frac{dy}{dx} = \frac{k}{y} \Rightarrow y dy = k dx$

$$\frac{y^2}{2} = kx + c \text{ passes through } (0,0)$$

$$\Rightarrow c = 0$$

$$\therefore y^2 = 2kx \text{ passes through } (1,2)$$

$$\text{curve is } y^2 = 4x.$$

$$\text{tangent to parabola is } y = mx + \frac{1}{m}.$$

$$\text{for common tangent } \frac{1}{m\sqrt{1+m^2}} = 1$$

$$\Rightarrow m^4 + m^2 - 1 = 0$$

$$m^2 = \frac{-1 \pm \sqrt{5}}{2}$$

$$m^2 = \frac{\sqrt{5}-1}{1} = 2 \sin 18^\circ$$

I03/3/115. If a circle touches the conic C at the point P(1,2) and passes through the focus of the conic then its radius is :

यदि एक वृत्त, शंकव C को बिन्दु P(1,2) पर स्पर्श करता है तथा शंकव की नाभि से गुजरता हो, तो वृत्त की त्रिज्या होगी :

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\sqrt{3}$

Ans. B

Sol. Equation of tangent to $y^2 = 4x$ at P(1,2) is
 $y = x + 1$.

$\therefore$ equation of required circle

$$(x-1)^2 + (y-2)^2 + \lambda(y-x-1) = 0$$

passes through (1,0) $\Rightarrow \lambda = 2$

$$\therefore \text{circle is } (x-1)^2 + (y-2)^2 + 2(y-x-1) = 0$$

$$\Rightarrow x^2 + y^2 - 4x + 2y + 3 = 0$$

$$\therefore \text{radius} = \sqrt{4+1-3} = \sqrt{2}$$

Paragraph for Question I03/3/116 to I03/3/118

Suppose $f(x)$ and $g(x)$ are differentiable functions such that $xg(f(x)) \cdot f'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(f(x)) f'(x) \forall x$

$\in \mathbb{R}$ and $f(x)$ & $g(x)$ are positive for all $x \in \mathbb{R}$. Also $\int_0^x f(g(t)) dt = \frac{1}{2}(1 - e^{-2x}) \forall x \in \mathbb{R}$, $g(f(0)) = 1$ and

$$h(x) = \frac{g(f(x))}{f(g(x))} \forall x \in \mathbb{R}.$$

माना $f(x)$ तथा $g(x)$ दो अवकलनीय फलन इस प्रकार हैं कि $xg(f(x)) \cdot f'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(f(x)) f'(x) \forall x \in \mathbb{R}$ तथा

x के सभी वास्तविक मानों के लिए दोनों $f(x)$ एवं $g(x)$ धनात्मक हैं और $\int_0^x f(g(t)) dt = \frac{1}{2}(1 - e^{-2x}) \forall x \in \mathbb{R}$, $g(f(0)) = 1$

$$\text{तथा } h(x) = \frac{g(f(x))}{f(g(x))} \forall x \in \mathbb{R}.$$

I03/3/116. The graph of $y = h(x)$ is symmetric with respect to the line :

निम्न में से किस रेखा के सापेक्ष $y = h(x)$ सममित है :

- (A) $x = -1$ (B) $x = 0$ (C) $x = 1$ (D) $x = 2$

Ans. C

Sol.

I03/3/117. The value $f(g(0)) + g(f(0))$ is equal to :

$f(g(0)) + g(f(0))$ का मान है :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol.

I03/3/118. The largest possible value of $h(x) \forall x \in \mathbb{R}$ is :

$h(x) \forall x \in \mathbb{R}$ का अधिकतम संभावित मान है :

- (A) 1 (B) $e^{\frac{1}{3}}$ (C) e (D) e^2

Ans. C

Sol.

Paragraph for Question I03/3/119 to I03/3/121

Let $y = f(x)$ satisfy the equation $f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t)dt$. Then,

माना $y = f(x)$ समीकरण $f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t)dt$ को संतुष्ट करता है, तब

I03/3/119. $f(x)$ satisfies the differential equation :

(A) $\frac{dy}{dx} + y = e^x (\cos x - \sin x) - e^{-x} (\cos x + \sin x)$

(B) $\frac{dy}{dx} = y + e^x (\cos x + \sin x) + e^{-x} (\cos x - \sin x)$

(C) $\frac{dy}{dx} + y = e^x (\cos x + \sin x) - e^{-x} (\cos x + \sin x)$

(D) None of these

$f(x)$ अवकल समीकरण को संतुष्ट करता है :

(A) $\frac{dy}{dx} + y = e^x (\cos x - \sin x) - e^{-x} (\cos x + \sin x)$

(B) $\frac{dy}{dx} = y + e^x (\cos x + \sin x) + e^{-x} (\cos x - \sin x)$

(C) $\frac{dy}{dx} + y = e^x (\cos x + \sin x) - e^{-x} (\cos x + \sin x)$

(D) इनमें से कोई नहीं

Ans. A

Sol.

I03/3/120. $f'(0) + f''(0) = :$

(A) -1

(B) 2

(C) 0

(D) 1

Ans. C1

Sol.

I03/3/121. $f(x)$ as a function of x equals :

(A) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x + \sin x) + \frac{2}{5} e^{-x}$

(B) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) - \frac{2}{5} e^{-x}$

(C) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) + \frac{2}{5} e^{-x}$

(D) None of these

$f(x)$ बराबर है :

(A) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x + \sin x) + \frac{2}{5} e^{-x}$

(B) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) - \frac{2}{5} e^{-x}$

(C) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) + \frac{2}{5} e^{-x}$

(D) इनमें से कोई नहीं

Ans. C

Sol.

Paragraph for Question I01/3/122 to I01/3/123

Consider two differentiable functions $f(x)$, $g(x)$ satisfying $6 \int f(x)g(x)dx = x^6 + 3x^4 + 3x^2 + c$ &

$$2 \int \frac{g(x)dx}{f(x)} = x^2 + c \text{ . where } f(x) > 0 \forall x \in \mathbb{R} \text{ .}$$

On the basis of above information, answer the following questions :

माना दो अवकलनीय फलन $f(x)$ तथा $g(x)$, $6 \int f(x)g(x)dx = x^6 + 3x^4 + 3x^2 + c$ तथा $2 \int \frac{g(x)dx}{f(x)} = x^2 + c$, जहाँ $f(x)$

$> 0 \forall x \in \mathbb{R}$ है, को संतुष्ट करते हैं।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I01/3/122. $f'(1) + g'(2)$ is equal to :

$f'(1) + g'(2)$ का मान होगा :

- (A) 15 (B) 14 (C) 13 (D) 12

Ans. A

Sol. Differentiating both sides we get

$$g(x) \cdot f(x) = (x^2 + 1)(x^3 + x)$$

$$\Rightarrow \frac{g(x)}{f(x)} = x$$

$$\Rightarrow g(x) = x^3 + x$$

$$f(x) = x^2 + 1$$

I01/3/123. $\int (f(x) + g(x))dx$ is equal to :

$\int (f(x) + g(x))dx$ का मान होगा :

- (A) $\frac{x^4}{4} + \frac{x^2}{2} + x + c$ (B) $\frac{x^4}{4} + \frac{x^2}{3} + x^2 + 2x + c$
(C) $\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x + c$ (D) $\frac{x^4}{4} + \frac{x^3}{3} + c$

Ans. C

Sol.

Paragraph for Question I01/3/124 to I01/3/125

$$\text{Let } 2I_1 + I_2 = \int \frac{e^x}{e^{2x} + e^{-2x}} dx \text{ and } I_1 + 2I_2 = \int \frac{e^{-x}}{e^{2x} + e^{-2x}} dx .$$

On the basis of above information, answer the following questions :

$$\text{माना } 2I_1 + I_2 = \int \frac{e^x}{e^{2x} + e^{-2x}} dx \text{ तथा } I_1 + 2I_2 = \int \frac{e^{-x}}{e^{2x} + e^{-2x}} dx \text{ है।}$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I01/3/124. I_1 is equal to :

I_1 बराबर होगा :

$$(A) \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$$

$$(B) \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$$

$$(C) \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x + e^{-x}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x - e^{-x} - \sqrt{2}}{e^x - e^{-x} + \sqrt{2}} \right| + C$$

$$(D) \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x + e^{-x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x - e^{-x} - \sqrt{2}}{e^x - e^{-x} + \sqrt{2}} \right| + C$$

Ans. A

$$\text{Sol. } 3I_1 + 3I_2 = \int \frac{e^x + e^{-x}}{(e^x - e^{-x})^2 + 2} dx$$

$$e^x - e^{-x} = t \quad \Rightarrow \quad (e^x + e^{-x}) dx = dt$$

$$3I_1 + 3I_2 = \int \frac{dt}{t^2 + 2} = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) + C$$

$$I_1 + I_2 = \frac{1}{3\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) + C \quad \dots(i)$$

$$\& \quad I_1 - I_2 = \int \frac{e^x - e^{-x}}{(e^x + e^{-x})^2 - 2} dx$$

$$e^x + e^{-x} = z \quad \Rightarrow \quad (e^x - e^{-x}) dx = dz$$

$$I_1 - I_2 = \int \frac{dz}{z^2 - 2} = \frac{1}{2\sqrt{2}} \ln \left| \frac{z - \sqrt{2}}{z + \sqrt{2}} \right| + C \quad \dots(ii)$$

From (i) & (ii)

$$I_1 = \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$$

&

$$I_2 = \frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$$

I01/3/125. I_2 is equal to :

I_2 बराबर होगा :

(A) $\frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$

(B) $\frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x - e^{-x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x + e^{-x} - \sqrt{2}}{e^x + e^{-x} + \sqrt{2}} \right| + C$

(C) $\frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x + e^{-x}}{\sqrt{2}} \right) + \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x - e^{-x} - \sqrt{2}}{e^x - e^{-x} + \sqrt{2}} \right| + C$

(D) $\frac{1}{6\sqrt{2}} \tan^{-1} \left(\frac{e^x + e^{-x}}{\sqrt{2}} \right) - \frac{1}{4\sqrt{2}} \ln \left| \frac{e^x - e^{-x} - \sqrt{2}}{e^x - e^{-x} + \sqrt{2}} \right| + C$

Ans.

B

Sol.

Paragraph for Question I01/3/126 to I01/3/127

Let a, b, c, d be 4 distinct roots of the equation $x^4 - 4x + 3 = x(x^3 - f'(1)x^2 + f''(1)x - 4) + f(1)$ and $f(x)$ is a monic polynomial of degree 3.

माना a, b, c, d समीकरण $x^4 - 4x + 3 = x(x^3 - f'(1)x^2 + f''(1)x - 4) + f(1)$ के चार भिन्न हल हैं तथा $f(x)$, त्रिघातीय इकाई मुख्य गुणांक वाला बहुपद है।

I01/3/126. The value of $\lim_{x \rightarrow 1} \frac{(x-1)^3 - \sin^3(x-1)}{(f(x)-3)^{5/3}}$ is :

$\lim_{x \rightarrow 1} \frac{(x-1)^3 - \sin^3(x-1)}{(f(x)-3)^{5/3}}$ का मान है :

(A) $\frac{1}{2}$

(B) 2

(C) $\frac{1}{6}$

(D) $\frac{2}{3}$

Ans. A**Sol.** $f'(1)x^3 + f''(1)x^2 + f(1) - 3 = 0$ has four distinct roots $\Rightarrow$ identity

Thus $f'(1) = 0, f''(1) = 0, f(1) = 3$

Gives

$f''(x) = a(x-1)$ (integrating)

$$f'(x) = \frac{a(x-1)^2}{2} + c$$

$$f'(x) = \frac{a(x-1)^3}{6} + cx + d$$

again integrating and finding constant using

$f'(1) = 0, f''(1) = 0, f(1) = 3$

We get

$f(x) = (x-1)^3 + 3$

$$g(x) = \tan^{-1}\left(\frac{x-1}{2}\right)$$

I01/3/127. $\int \frac{dx}{\frac{f(x)-3}{x-1} + 4} = \frac{1}{2}g(x) + C$, where C is constant of integration and $g(3) = \frac{\pi}{4}$, then the value of $g(5) + g(7)$

is:

$$\int \frac{dx}{\frac{f(x)-3}{x-1} + 4} = \frac{1}{2}g(x) + C, \text{ जहाँ } C \text{ समाकलन नियतांक है तथा } g(3) = \frac{\pi}{4}, \text{ तब } g(5) + g(7) \text{ का मान होगा :}$$

(A) $\frac{\pi}{2}$

(B) π

(C) $\frac{5\pi}{4}$

(D) $\frac{3\pi}{4}$

Ans. D**Sol.** $f'(1)x^3 + f''(1)x^2 + f(1) - 3 = 0$ has four distinct roots $\Rightarrow$ identity

Thus $f'(1) = 0, f''(1) = 0, f(1) = 3$

Gives

$f''(x) = a(x-1)$ (integrating)

$$f'(x) = \frac{a(x-1)^2}{2} + c$$

$$f'(x) = \frac{a(x-1)^3}{6} + cx + d$$

again integrating and finding constant using

$$f'(1) = 0, f''(1) = 0, f(1) = 3$$

We get

$$f(x) = (x-1)^3 + 3$$

$$g(x) = \tan^{-1}\left(\frac{x-1}{2}\right)$$

Paragraph for Question I01/3/128 to I01/3/130

Let f, g, h be 3 functions such that $f'(x) > 0$ & $g(x) > 0 \forall x \in \mathbb{R}$ where $\int f(x).g(x)dx = \frac{x^4}{4} + C$ &

$$\int \frac{f(x)}{g(x)} dx = \int \frac{g(x)}{h(x)} dx = \ln|x| + C, x \neq 0. \text{ On the basis of above information, answer the following questions}$$

:

माना $x \in \mathbb{R}$ के लिए f, g, h तीन फलन इस प्रकार हैं कि $f'(x) > 0$ तथा $g(x) > 0$ है, जहाँ $\int f(x).g(x)dx = \frac{x^4}{4} + C$ तथा

$$\int \frac{f(x)}{g(x)} dx = \int \frac{g(x)}{h(x)} dx = \ln|x| + C, x \neq 0 \text{ है।}$$

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उत्तर दीजिए। :

I01/3/128. $\int f(x).g(x).h(x)dx$ is equal to :

$\int f(x).g(x).h(x)dx$ का मान होगा :

$$(A) \frac{x^5}{5} + C \quad (B) \frac{x^6}{6} + C \quad (C) \frac{x^7}{7} + C \quad (D) \frac{x^8}{8} + C,$$

Ans. C

Sol. $f'(x) > 0$ and $g(x) > 0$ and $x \in \mathbb{R}$

$$\int f(x).g(x)dx = \frac{x^4}{4} + c \text{ and } \int \frac{f(x)}{g(x)} dx = \int \frac{g(x)}{h(x)} dx = \ln|x| + c$$

$$f(x)g(x) = x^3 \text{ and } \frac{f(x)}{g(x)} = \frac{g(x)}{h(x)} = \frac{1}{x}$$

$$g(x) = xf(x), h(x) = g(x)$$

$$h(x) = x^2 f(x)$$

$$\therefore f(x).g(x) = x^3$$

$$(f(x))(x f(x)) = x^3$$

$$f^2(x) = x^2$$

$$f(x) = \pm x$$

$$\therefore f'(x) > 0 \Rightarrow f(x) = x$$

$$\Rightarrow g(x) = x^2 \text{ and } h(x) = x^3$$

$$\int f(x).g(x).h(x)dx = \int x_1 x^2 . x^3 dx = \frac{x^7}{7} + c$$

I01/3/129. $f(x), g(x), h(x)$ are in :

- (A) A.P. (B) G.P. (C) H.P. (D) None of these

$f(x), g(x), h(x)$ होंगें :

- (A) समान्तर श्रेणी में (B) गुणोत्तर श्रेणी में (C) हरात्मक श्रेणी में (D) इनमें से कोई नहीं

Ans. B

Sol. $f(x), g(x), h(x) = x, x^2, x^3 \Rightarrow G.P.$

I01/3/130. $\int \frac{h(x)dx}{\sqrt{1+g(x)}}$ is equal to :

$\int \frac{h(x)dx}{\sqrt{1+g(x)}}$ का मान होगा :

(A) $\frac{\sqrt{x^2+1}(x^2-2)}{3} + C$

(B) $\frac{\sqrt{x^2+1}(x^2-1)}{3} + C$

(C) $\frac{\sqrt{x^2+1}(x^2+1)}{3} + C$

(D) $\frac{\sqrt{x^2+1}(x^2-3)}{3} + C$

Ans. A

Sol. $\int \frac{h(x)}{\sqrt{1+g(x)}} dx = \int \frac{x^3}{\sqrt{1+x^2}} dx$ Let $1+x^2 = t$ $2x dx = dt$

$$\begin{aligned}
&= \int \frac{x^2 \cdot x}{\sqrt{1+x^2}} dx \\
&= \int \frac{(t-1)}{\sqrt{t}} \times \frac{dt}{2} \\
&= \frac{1}{2} \left[\sqrt{t} - \frac{1}{\sqrt{t}} \right] dt = \frac{1}{2} \left[\frac{2t^{\frac{3}{2}}}{3} - \frac{2t^{\frac{1}{2}}}{1} \right] + c \\
&= \sqrt{t} \left[\frac{t-3}{3} \right] = \sqrt{1+x^2} \left(\frac{x^2-2}{3} \right) + c
\end{aligned}$$

Paragraph for Question I01/3/131 to I01/3/133

Consider, $\int (\cos^{-1} x + \cos^{-1} \sqrt{1-x^2}) dx = Ax + f(x) \sin^{-1} x + C\sqrt{1-x^2} + D \quad \forall x \in [-1, 0)$

On the basis of above information answer the following :

माना, $\int (\cos^{-1} x + \cos^{-1} \sqrt{1-x^2}) dx = Ax + f(x) \sin^{-1} x + C\sqrt{1-x^2} + D \quad \forall x \in [-1, 0)$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए। :

I01/3/131. The value of A + C is :

A + C का मान होगा :

- (A) $\pi + 1$ (B) $-\frac{\pi}{2} + 2$ (C) $\frac{\pi}{2} - 2$ (D) $-\pi - 1$

Ans. C

Sol. $\int (\cos^{-1} x + \cos^{-1} \sqrt{1-x^2}) dx = Ax + f(x) \sin^{-1} x + C$

put $x = \cos \theta \Rightarrow \theta = \cos^{-1} x$

$$\because x \in [-1, 0) \Rightarrow \theta \left(\frac{\pi}{2}, \pi \right]$$

$$= - \int \left[\cos^{-1}(\cos \theta) + \cos^{-1} \left(\cos \left(\frac{\pi}{2} - \theta \right) \right) \right] \sin \theta d\theta$$

$$= - \int \left[\theta + 0 - \left(\frac{\pi}{2} - \theta \right) \right] \sin \theta d\theta$$

$$\begin{aligned}
&= -\int \left(2\theta - \frac{\pi}{2}\right) \sin \theta d\theta \\
&= -2 \int \theta \sin \theta d\theta + \frac{\pi}{2} \int \sin \theta d\theta \\
&= +2 [\theta \cos \theta - \sin \theta] - \frac{\pi}{2} \cos \theta + D \\
&= 2 \left[(\cos^{-1} x)(x) - \sqrt{1-x^2} \right] - \frac{\pi}{2} x + D \\
&= 2 \left[\left(\frac{\pi}{2} - \sin^{-1} x \right) x - \sqrt{1-x^2} \right] - \frac{\pi}{2} x + D \\
&= \frac{\pi}{2} x - 2x \sin^{-1} x - 2\sqrt{1-x^2} + D \\
A &= \frac{\pi}{2}, f(x) = -2x, C = -2, D
\end{aligned}$$

I01/3/132. If $\int f(x)dx = g(x)$ and $g\left(-\frac{1}{2}\right) = 0$, then number of solutions of the equation $g(x) = \text{sgn}(\sin^{-1}x)$, where $\text{sgn}(\cdot)$ denotes signum function :

यदि $\int f(x)dx = g(x)$ तथा $g\left(-\frac{1}{2}\right) = 0$ है, तो समीकरण $g(x) = \text{sgn}(\sin^{-1}x)$ के हलों की संख्या, जहाँ $\text{sgn}(\cdot)$ सिग्नम

फलन को दर्शाता है, होगी :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\int (-2x)dx = g(x)$

$$-x^2 + C = g(x)$$

$$x^2 + \frac{1}{4} = g(x)$$

$$\frac{1}{4} - x^2 = \text{sgn}(\sin^{-1} x)$$

$$\because x \in [-1, 0) \Rightarrow \sin^{-1} x \in \left[-\frac{\pi}{2}, 0\right)$$

$$\frac{1}{4} - x^2 = -1 \Rightarrow x = \pm \frac{\sqrt{5}}{2} \Rightarrow \text{No solution}$$

I01/3/133. If range of $f(x)$ is $(a, b]$ and $\int \sin^{-1}(\cos x) dx = K.x + Fx^2 + G, \forall x \in (a, b]$, then $[K], [3 + F], [K - C + 1]$, where $[.]$ is greatest integer function, are in :

- (A) A.P. (B) G.P. (C) H.P. (D) none of these

यदि $f(x)$ का परिसर $(a, b]$ तथा $\int \sin^{-1}(\cos x) dx = K.x + Fx^2 + G, \forall x \in (a, b]$ है, तो $[K], [3 + F], [K - C + 1]$, जहाँ $[.]$ महतम पूर्णांक फलन है, होंगे :

- (A) समान्तर श्रेणी (B) गुणोत्तर श्रेणी (C) हरात्मक श्रेणी (D) इनमें से कोई नहीं

Ans. B

Sol. $\int \sin^{-1}(\cos x) dx$

$$= \int \left(\frac{\pi}{2} - (\cos^{-1}(\cos x)) \right) dx$$

$$= \int \left(\frac{\pi}{2} - x \right) dx \quad \{ \because \cos^{-1}(\cos x) = x \forall x \in (0, 2] \}$$

$$= \frac{\pi}{2}x - \frac{x^2}{2} + G$$

$$\therefore K = \frac{\pi}{2}, F = -\frac{1}{2}, C = -2$$

$$[K] = 1, [3 + F] = 2, [K - C + 1] = \left[\frac{\pi}{2} + 2 + 1 \right] = 4$$

Hence, $[K], [3 + F], [K - C + 1]$ form a G.P.

Paragraph for Question I01/3/134 to I01/3/136

Let $F(x)$ be a differentiable function such that $\frac{d}{dx}(F(x)) = \frac{e^x}{x}$ where curve of $y = F(x)$ passes through

$$\left(\pm \frac{1}{2}, 0 \right) \text{ and } G(x) = \frac{1}{e} \int \frac{(2x^3 - x)e^{x^2} dx}{x^2(x^2 - 1)} \text{ \& } G\left(\frac{1}{\sqrt{2}}\right) = 0.$$

On the basis of above information, answer the following questions :

माना $F(x)$ एक अवकलनीय फलन इस प्रकार है कि $\frac{d}{dx}(F(x)) = \frac{e^x}{x}$ जहाँ वक्र $y = F(x)$ बिन्दु $\left(\pm \frac{1}{2}, 0\right)$ से गुजरता है तथा

$$G(x) = \frac{1}{e} \int \frac{(2x^3 - x)e^{x^2} dx}{x^2(x^2 - 1)} \text{ व } G\left(\frac{1}{\sqrt{2}}\right) = 0 \text{ है।}$$

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I01/3/134. $\lim_{x \rightarrow 0} \frac{G'(x)}{F'(x)}$ is (where $G'(x)$ and $F'(x)$ are derivatives of $G(x)$ and $F(x)$ w.r.t. x respectively) :

$\lim_{x \rightarrow 0} \frac{G'(x)}{F'(x)}$ का मान होगा (जहाँ $G'(x)$ तथा $F'(x)$ क्रमशः $G(x)$ तथा $F(x)$ के x के सापेक्ष अवकलज को दर्शाता है) :

- (A) 1 (B) $-\frac{1}{e}$ (C) $\frac{1}{e}$ (D) -1

Ans. C

Sol.
$$\lim_{x \rightarrow 0} \frac{\frac{1}{e} \frac{(2x^3 - x)e^{x^2}}{x^2(x^2 - 1)}}{\frac{e^x}{x}} = \lim_{x \rightarrow 0} \left(\frac{1}{e} \right) \left[\frac{(2x^2 - 1)}{(x^2 - 1)} \cdot e^{x^2 - x} \right] = \frac{1}{e}$$

I01/3/135. If $F(1) = F(2) \neq 0$, then $\frac{G(-\sqrt{2})}{F(2)}$ is equal to :

यदि $F(1) = F(2) \neq 0$ हो, तो $\frac{G(-\sqrt{2})}{F(2)}$ का मान होगा :

- (A) $\frac{1-e}{2e}$ (B) $-\frac{1-e}{2e}$ (C) $\frac{e+1}{2e}$ (D) $\frac{e-1}{2e}$

Ans. C

Sol.
$$\therefore F'(x) = \frac{e^x}{x} \text{ and } G'(x) = \frac{1}{e} \frac{(2x^3 - x)e^{x^2}}{x^2(x^2 - 1)}$$

$$\therefore \lim_{x \rightarrow 0} \frac{G'(x)}{F'(x)} = \lim_{x \rightarrow 0} \frac{1}{e} \frac{(2x^3 - x)e^{x^2}}{x^2(x^2 - 1)} \times \frac{x}{e^x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{e} \frac{(2x^2 - 1)e^{x^2}}{(x^2 - 1)e^x} = \frac{1}{e}$$

$$\text{Put } x = -\sqrt{2}$$

$$\Rightarrow G(-\sqrt{2}) = \frac{1}{2} \left[\frac{F(2)}{e} + F(1) \right] = \frac{1+e}{2e} F(1)$$

$$\Rightarrow \frac{G(-\sqrt{2})}{F(1)} = \frac{1+e}{2e}$$

$$G(x) = \frac{1}{e} \int \frac{(2x^2 - 1)x \cdot e^{x^2} dx}{x^2(x^2 - 1)}$$

$$\text{Put } x^2 = t \quad \Rightarrow \quad x dx = \frac{dt}{2}$$

$$= \frac{1}{e} \int \frac{2t-1}{t(t-1)} \cdot e^t \frac{dt}{2} = \frac{1}{2e} \int \left\{ \frac{1}{t} + \frac{1}{t-1} \right\} \cdot e^t dt$$

I01/3/136. $G(x)$ is equal to :

$G(x)$ बराबर होगा :

$$(A) \frac{1}{2} \left[eF(x^2) + F(x^2 - 1) \right]$$

$$(B) \frac{1}{2e} \left[eF(x^2) + F(x^2 - 1) \right]$$

$$(C) \frac{1}{2e} \left[eF(x^2) + F(x^2 - 1) \right]$$

$$(D) \frac{1}{2e} \left[eF(x^2) + F(x^2 - 1) \right]$$

Ans. D

$$\begin{aligned} \text{Sol.} \quad &= \frac{1}{2e} \int \frac{e^t}{t} dt + \frac{1}{2e} \int \frac{e^t}{t-1} dt \\ &= \frac{1}{2e} \cdot F(t) + \frac{1}{2} \cdot F(t-1) + C \left(\because \frac{d}{dx} (F(x)) = \frac{e^x}{x} \right) \end{aligned}$$

$$\Rightarrow G(x) = \frac{1}{2} \left[\frac{1}{e} F(x^2) + F(x^2 - 1) \right] + C$$

$$\text{Put } x = \frac{1}{\sqrt{2}} \Rightarrow G\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2} \left[\frac{1}{e} \cdot F\left(\frac{1}{2}\right) + F\left(-\frac{1}{2}\right) \right] + C$$

$$\therefore F\left(\pm \frac{1}{2}\right) = 0 \text{ and } G\left(\frac{1}{\sqrt{2}}\right) = 0$$

$$\therefore C = 0$$

$$\Rightarrow G(x) = \frac{1}{2} \left[\frac{F(x^2)}{e} + F(x^2 - 1) \right]$$

INTEGER TYPE QUESTIONS

I04/4/137. Area of the region bounded by $[x]^2 = [y]^2$, if $x \in [1, 5]$, where $[.]$ denotes the greatest integer function, is _____.

$x \in [1, 5]$ में $[x]^2 = [y]^2$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है, जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है _____.

Ans. 8

Sol. $[x]^2 = [y]^2$

$$\therefore [y] = \pm [x]$$

$$[y] = \pm 1, \quad 1 \leq x < 2$$

$$= \pm 2, \quad 2 \leq x < 3$$

$$= \pm 3, \quad 3 \leq x < 4$$

$$= \pm 4, \quad 4 \leq x < 5$$

$$= \pm 5, \quad x = 5$$

Now, when, $x \in [1, 2)$

$$\text{then } y \in [-1, 0) \cup [1, 2)$$

$$\text{when } x \in [2, 3)$$

$$\text{then } y \in [-2, -1) \cup [2, 3)$$

$$\text{when, } x \in [3, 4)$$

$$\text{then } y \in [-3, -2) \cup [3, 4)$$

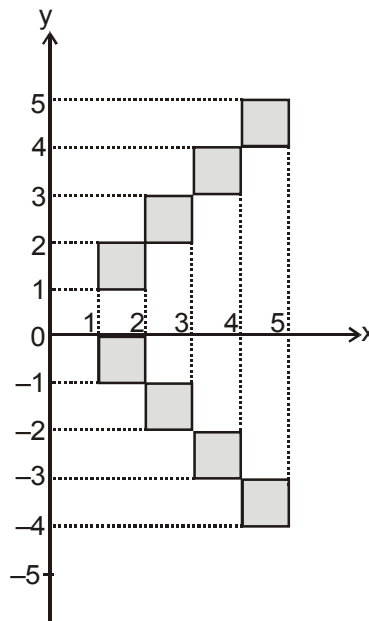
$$\text{when, } x \in [4, 5)$$

$$\text{then } y \in [-4, -3) \cup [4, 5)$$

$$\text{when, } x = 5$$

$$\text{then } y \in [-5, -4) \cup [5, 6)$$

Hence, required area = $2(4) = 8$ sq unit



I04/4/138. Consider $y = x^2$ and $f(x)$ where $f(x)$ is a differentiable function satisfying $f(x+1) + f(z-1) = f(x+z) \forall x,$

$z \in \mathbb{R}$ and $f(0) = 0$; $f'(0) = 4$. If area bounded by curve $y = x^2$ and $y = f(x)$ is Δ find the value of $\left(\frac{3}{16}\Delta\right)$

_____.

$y = x^2$ तथा $f(x)$ पर विचार कीजिए जहाँ $f(x), f(x+1) + f(z-1) = f(x+z) \forall x, z \in \mathbb{R}$ को संतुष्ट करने वाला एक अवकलनीय फलन है तथा $f(0) = 0; f'(0) = 4$ हैं। यदि वक्र $y = x^2$ तथा $y = f(x)$ द्वारा परिबद्ध क्षेत्रफल Δ है। $\left(\frac{3}{16}\Delta\right)$ का मान ज्ञात कीजिए _____.

Ans. 2

Sol. $f(x) + f(z) = f(x+z)$ and $f(0) = 0$ and $f'(0) = 4 \Rightarrow f(x) = 4x$

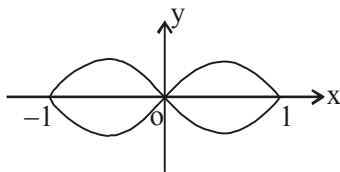
$$\text{So area bounded} = \Delta = \int_0^4 (4x - x^2) dx$$

I04/4/139. The least integer which is greater than or equal to the area of region in $x-y$ plane satisfying $x^6 - x^2 + y^2 \leq 0$ is _____.

न्यूनतम पूर्णांक जो xy समतल में $x^6 - x^2 + y^2 \leq 0$ को संतुष्ट करने वाले क्षेत्र के क्षेत्रफल से बड़ा या बराबर है, होगा _____.

Ans. 2

Sol. Required area $= 4 \int_0^1 \sqrt{x^2 - x^6} dx = \int_0^1 4x\sqrt{1-x^4} dx = \frac{\pi}{2}$



Put $x^2 = \sin\theta$

I04/4/140. Area of the smaller region bounded by $|y| = 4 - x^2$ and $|y| = 3x$ can be expressed as p/q sq. units. Where p and q are relatively prime positive integer find the value of $(p+q)$. _____.

$|y| = 4 - x^2$ तथा $|y| = 3x$ द्वारा परिबद्ध लघु क्षेत्र के क्षेत्रफल को p/q वर्ग इकाई द्वारा दर्शाया जाता है जहाँ p तथा q परस्पर अभाज्य धनात्मक पूर्णांक हैं। $(p+q)$ का मान ज्ञात कीजिए _____.

Ans. 22

Sol. $|y| = 4 - x^2$

$$|y| = 3x$$

$$3x = 4 - x^2$$

$$x^2 + 3x - 4 = 0$$

$$x^2 + 4x - x - 4$$

$$(x+4)(x-1) = 0$$

$$x = 1$$

$$\text{are (OABCO)} = 2(\text{area OABO}) = 2 \left[\int_0^1 3x \, dx + \int_1^2 (4 - x^2) \, dx \right] = 2 \left\{ \left(\frac{3}{2} \right) + \left(4 - \frac{1}{3}(2^3 - 1) \right) \right\}$$

$$= 2 \left\{ \frac{3}{2} + \frac{12-7}{3} \right\} = 2 \left\{ \frac{3}{2} + \frac{5}{3} \right\} = \frac{19}{3} = \frac{p}{q}$$

$$\therefore p + q = 22$$

I04/4/141. The common tangents to curves $y^2 = 8x$ and $15x^2 + 4y^2 = 60$ enclose an area 2^k sq. units with y-axis. Value of k is :

वक्रों $y^2 = 8x$ तथा $15x^2 + 4y^2 = 60$ की उभयनिष्ठ स्पर्श रेखाओं तथा y-अक्ष द्वारा परिबद्ध क्षेत्रफल 2^k वर्ग इकाई है। k का मान होगा :

Ans. 5

Sol. Tangent to parabola $y^2 = 8x$ and $y = mx + \frac{2}{m}$

$$\therefore \text{this is tangent to } \frac{x^2}{4} + \frac{y^2}{15} = 1$$

$$\Rightarrow \left(\frac{2}{m} \right)^2 = 4m^2 + 15$$

$$\Rightarrow m = \pm \frac{1}{2}$$

$$\therefore \text{tangents : } x \pm 2y + 8 = 0$$

Now get area = 32 sq. units.

I04/4/142. A curve passes through $\left(1, \frac{5}{2}\right)$ and slope of tangent at point P(x, y) equals $\frac{y + x^3 + 2x^2}{x}$. If the area

enclosed by the curve and x-axis in second quadrant is A, then $\frac{3}{4}A$ is :

वक्र जो बिन्दु $\left(1, \frac{5}{2}\right)$ से गुजरता है तथा वक्र के बिन्दु P(x, y) पर स्पर्श रेखा की प्रवणता $\frac{y + x^3 + 2x^2}{x}$ है यदि वक्र तथा x-

अक्ष द्वारा द्वितीय चतुर्थांश में परिबद्ध क्षेत्रफल A है तो $\frac{3}{4}A$ का मान होगा :

Ans. 8

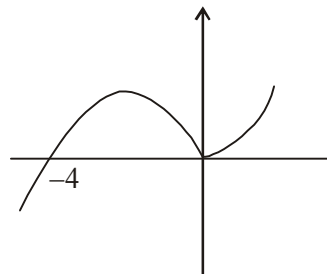
Sol. $\frac{dy}{dx} = \frac{y + x^3 + 2x^2}{x} \Rightarrow \frac{dy}{dx} - \frac{y}{x} = x^2 + 2x$

$$\Rightarrow y \frac{1}{x} = \int (x+2) dx$$

$$\Rightarrow \frac{y}{x} = \frac{x^2}{2} + 2x + C \text{ \& } C = 0 \Rightarrow y = \frac{x^2}{2} (x+4)$$

$$A = \int_{-4}^0 \frac{x^2}{2} (x+4) = \left(\frac{x^4}{8} + \frac{2}{3} x^3 \right)_{-4}^0$$

$$= 0 - \left[32 - \frac{128}{3} \right] = \frac{32}{3} \Rightarrow \frac{3}{4} A = 8$$



I04/4/143. Area of the region in $x-y$ plane such that $\sin^2 x - \sin x \sin y + \sin^2 y \leq \frac{3}{4}$, where $x, y \in \left[0, \frac{\pi}{2}\right]$, is $\frac{\pi^2}{k}$, then the value of k is.....

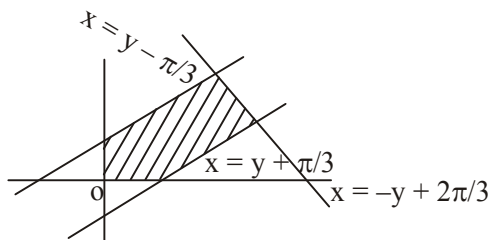
$x-y$ तल में क्षेत्र इस प्रकार है कि $\sin^2 x - \sin x \sin y + \sin^2 y \leq \frac{3}{4}$, जहाँ $x, y \in \left[0, \frac{\pi}{2}\right]$, है तथा इस क्षेत्र का क्षेत्रफल

$\frac{\pi^2}{k}$ है तो k का मान होगा :

Ans. 6

Sol. If $\sin^2 x - \sin x \sin y + \sin^2 y = \frac{3}{4}$

$$\Rightarrow \sin x = \frac{\sin y}{2} \pm \frac{\sqrt{3} \cos y}{2} = \sin \left(y \pm \frac{\pi}{3} \right)$$



I04/4/144. If the area bounded by $2^{|x|}|y| + 2^{|x|-1} \leq 1$ is $\frac{a}{\ln 2} - b$ and the area bounded by $2^{|x|}|y| + 2^{|x|-1} \leq 1, |x| \leq \frac{1}{2}$ &
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$|y| \leq \frac{1}{2}$ is $\frac{(\sqrt{2}-1)\sqrt{c}}{\ln 2} - 1$, then find the value of $\frac{c}{a^2b}$ _____.

यदि $2^{|x|}|y| + 2^{|x|-1} \leq 1$ द्वारा परिबद्ध क्षेत्रफल $\frac{a}{\ln 2} - b$ तथा $2^{|x|}|y| + 2^{|x|-1} \leq 1, |x| \leq \frac{1}{2}$ द्वारा परिबद्ध क्षेत्रफल

$\frac{(\sqrt{2}-1)\sqrt{c}}{\ln 2} - 1$ है, तो $\frac{c}{a^2b}$ का मान ज्ञात करो।

Ans. 1

Sol. $|y| \leq 2^{-|x|} - \frac{1}{2}$

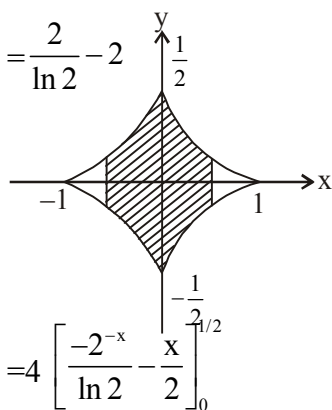
$$\text{Area} = 4 \int_0^1 \left(2^{-x} - \frac{1}{2} \right) dx$$

$$= 4 \left[\frac{-2^{-x}}{\ln 2} - \frac{x}{2} \right]_0^1$$

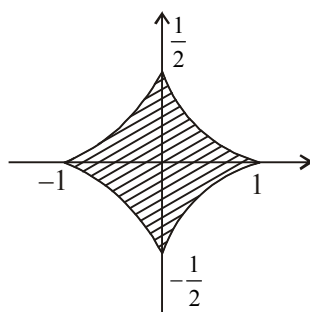
$$= 4 \left[\frac{1}{2 \ln 2} - \frac{1}{2} + \frac{1}{\ln 2} \right]$$

$$= 4 \left[\frac{1}{2 \ln 2} - \frac{1}{2} \right]$$

$$= \frac{2}{\ln 2} - 2$$



$$\text{Area} = 4 \left[\int_0^{\frac{1}{2}} 2^{-x} - \frac{1}{2} dx \right]$$



I04/4/145. Let S_1 and S_2 be the set of points in x-y plane (where $x \geq 0$ and $y \in \mathbb{R}$) defined as

$$S_1 = \{(x, y) : |x + |x|| + |y + |y|| \leq 2\}$$

$$S_2 = \{(x, y) : |x - |x|| + |y - |y|| \leq 2\}$$

If the area of intersection of S_1 and S_2 is A , then find the value of $2A$.

माना S_1 तथा S_2 , x - y समतल में बिन्दुओं के समुच्चय (जहाँ $x \geq 0$ तथा $y \in \mathbb{R}$) इस कार परिभाषित है कि

$$S_1 = \{(x, y) : |x + |x|| + |y + |y|| \leq 2\}$$

$$S_2 = \{(x, y) : |x - |x|| + |y - |y|| \leq 2\}$$

S_1 का S_2 के सर्वनिष्ठ का क्षेत्रफल A है, तो $2A$ का मान ज्ञात कीजिये।

Ans. 3

Sol. **Case-I:** $x \geq 0$ and $y \geq 0$

$$S_1 : 2x - 2y \leq 2 \Rightarrow x + y \leq 1 \text{ and}$$

$$S_2 : \text{True}$$

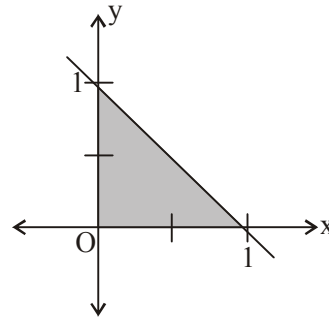
$$A = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

Case-II: $x \geq 0$ and $y \leq 0$

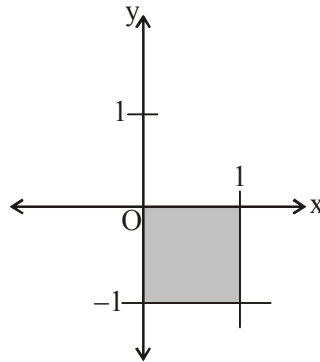
$$S_1 : 2x \leq 2 \Rightarrow x \leq 1$$

$$S_2 : -2y \leq 2 \Rightarrow y \geq -1$$

$$\text{Area} = 1.$$



$$\therefore \text{Area} = \frac{1}{2} + 1 = \frac{3}{2} = A \Rightarrow 2A = 3$$



I04/4/146. Area bounded by the curve $f(x) = \sqrt{\left(\frac{1}{\cos^4[x]} + [\sin x]\right)} - \tan^2[x]$, (where $[.]$ denotes greatest integer

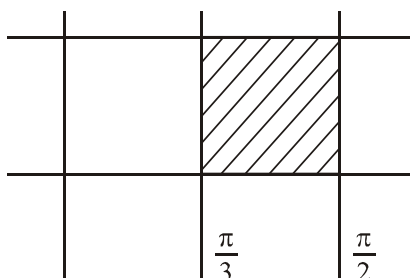
function) and the lines $x = \frac{\pi}{3}$ to $x = \frac{\pi}{2}$ and x -axis is $\frac{k\pi}{12}$, find the value of k .

वक्र $f(x) = \sqrt{\left(\frac{1}{\cos^4[x]} + [\sin x]\right)} - \tan^2[x]$, रेखा $x = \frac{\pi}{3}$ व $x = \frac{\pi}{2}$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल $\frac{k\pi}{12}$ है, तो k का

मान होगा (जहाँ $[\cdot]$ महत्तम पूर्णांक फलन को दर्शाता है)

Ans. 2

Sol.



$$\text{Area} = \left(\frac{\pi}{2} - \frac{\pi}{3}\right) 1 = \frac{\pi}{6}$$

I04/4/147. If the area bounded by the curve $y = |\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|$ and x-axis from $\frac{3\pi}{2} \leq x \leq 2\pi$, is equal to $\frac{\pi^2}{k}$, where $k \in \mathbb{N}$, then find k.

वक्र $y = |\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|$ तथा x-अक्ष $\frac{3\pi}{2} \leq x \leq 2\pi$ द्वारा परिबद्ध क्षेत्रफल $\frac{\pi^2}{k}$, $k \in \mathbb{N}$ है तो k का मान होगा :

Ans. 4

Sol. $y = |\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|; \frac{3\pi}{2} \leq x \leq 2\pi$

$$y = \frac{\pi}{2} - \sin^{-1}(\sin x) - \left| \frac{\pi}{2} - \cos^{-1}(\cos x) \right|$$

$$= \frac{\pi}{2} - (x - 2\pi) - \left| \frac{\pi}{2} - (2\pi - x) \right|$$

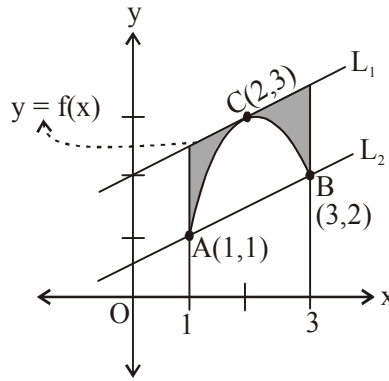
$$= \frac{\pi}{2} - x + 2\pi - \left| \frac{\pi}{2} - 2\pi + x \right|$$

$$= \frac{5\pi}{2} - x + \frac{3\pi}{2} - x$$

$$y = -2x + 4\pi$$

I04/4/148. The following figure shows the graph of a continuous function $y = f(x)$ on the interval $[1, 3]$. The points A, B, C have co-ordinates $(1, 1)$, $(3, 2)$, $(2, 3)$ respectively and the lines L_1 and L_2 are parallel with L_1 being tangents to the curve at C. If the area under the graph of $y = f(x)$ from $x = 1$ to $x = 3$ is 4 square units, then find the area (in square units) of shaded region.

दिये गये आरेख में अन्तराल $[1, 3]$ में फलन $y = f(x)$ सतत है बिन्दु A, B, C जिनके निर्देशांक क्रमशः $(1, 1)$, $(3, 2)$, $(2, 3)$ है। रेखाएँ L_1 तथा L_2 समान्तर है जिसमें रेखा L_1 वक्र C की स्पर्श रेखा है। यदि आरेख $y = f(x)$ का $x = 1$ से $x = 3$ द्वारा परिबद्ध क्षेत्रफल 4 वर्ग इकाई है तो छायांकित भाग का क्षेत्रफल (वर्ग इकाई में) होगा।

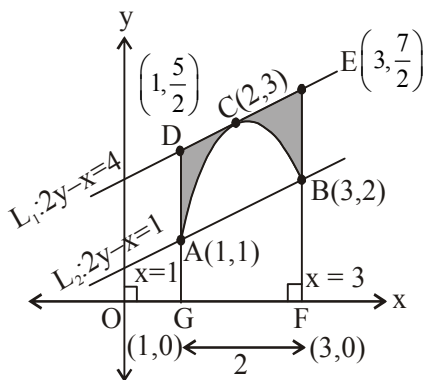


Ans. 2

Sol. $\therefore$ Area of shaded region

$$= (\text{area of trapezium DEFG}) - (\text{area under } f(x))$$

$$= \frac{1}{2} \left(\frac{5}{2} + \frac{7}{2} \right) \times 2 - 4$$



$$= 6 - 4 = 2$$

I02/4/149. If $P(t) = \int_0^{\infty} \frac{\ln y}{y^2 + ty + t^2} dy (t > 0)$ and $tP(t) - P(1) = \pi(3)^{-1/2}$. Also $t = \sqrt{e^\alpha}$, then the value of α is
_____.

यदि $P(t) = \int_0^{\infty} \frac{\ln y}{y^2 + ty + t^2} dy (t > 0)$ तथा $tP(t) - P(1) = \pi(3)^{-1/2}$, $t = \sqrt{e^\alpha}$ है, तो α का मान ज्ञात करो।

Ans. $\text{Int} = \frac{3}{2}$ or $t = e^{3/2}$

Sol. $P(1) = \int_0^1 \frac{\ln y}{y^2 + y + 1} dy + \int_1^{\infty} \frac{\ln y}{y^2 + y + 1} dy$

Put $y = \frac{1}{k}$ in $S_1 = \int_1^{\infty} \frac{\ln y}{y^2 + y + 1} dy$

We get $S_1 = -\int_0^1 \frac{\ln y}{y^2 + y + 1} dy \Rightarrow P(1) = 0$

in $P(t)$, put $y = \frac{t}{k}$

$$P(t) = \frac{1}{t} \int_0^{\infty} \frac{\ln t - \ln k}{k^2 + k + 1} dk$$

$$tP(t) = \ln t \int_0^{\infty} \frac{dk}{k^2 + k + 1} = \frac{\pi}{\sqrt{3}}$$

$$\Rightarrow \ln t = \frac{3}{2} \text{ or } t = e^{\frac{3}{2}}$$

I02/4/150. If the value of definite integral $\int_0^{\frac{\pi}{2}} \frac{1 - \sin 2x}{(1 + \sin 2x)^2} dx$ is equal to $\frac{m}{n}$ (where m and n are coprime), then
find the value of $(m + n)$. _____.

यदि निश्चित समाकल $\int_0^{\frac{\pi}{2}} \frac{1 - \sin 2x}{(1 + \sin 2x)^2} dx$ का मान $\frac{m}{n}$ है (जहाँ m तथा n सहअभाज्य संख्या है), तो $(m + n)$ का मान ज्ञात करो।

Ans. 4

Sol.

$$\int_0^{\frac{\pi}{2}} \underbrace{(\cos x - \sin x)}_I \cdot \underbrace{\frac{\cos x - \sin x}{(\sin x + \cos x)^4}}_{II} dx$$

$$= (\cos x - \sin x) \left(\frac{-1}{3(\sin x + \cos x)^3} \right) \Bigg|_0^{\frac{\pi}{2}}$$

$$= -\frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{(\sin x + \cos x)}{(\sin x + \cos x)^3} dx$$

$$= \frac{2}{3} - \frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{(\sec^2 x)}{(1 + \tan x)^2} dx$$

$$= \frac{1}{3} = \frac{m}{n}$$

$m + n = 4$

I02/4/151. Let $\int_0^1 \frac{4x^3(1+(x^4)^{2010})}{(1+x^4)^{2012}} dx = \frac{\lambda}{\mu}$

where λ and μ are relatively prime positive integers. Find unit digit of μ . _____.

माना $\int_0^1 \frac{4x^3(1+(x^4)^{2010})}{(1+x^4)^{2012}} dx = \frac{\lambda}{\mu}$

जहाँ λ तथा μ अभाज्य धनात्मक पूर्णांक हैं। μ का इकाई अंक ज्ञात कीजिए।

Ans. $\frac{\lambda}{\mu}$

Sol. Let $x^4 = t$
 $4x^3 dx = dt$

$$\int_0^1 \frac{1+t^{2010}}{(1+t)^{2012}} dt = \int_0^1 \frac{1}{(1+t)^{2012}} dt + \int_0^1 \frac{1}{t^2(1+1/t)^{2012}} dt = \frac{(1+t)^{-2011}}{-2011} \Big|_0^1 + \frac{(1+1/t)^{-2011}}{2011} \Big|_0^1$$

$$\frac{-1}{2011} \left(\frac{1}{2^{2011}} - 1 \right) + \frac{1}{2011} \left(\frac{1}{2^{2011}} - 0 \right) = \frac{-1}{2011} \left(\frac{1}{2^{2011}} - 1 - \frac{1}{2^{2011}} \right) = \frac{1}{2011} = \frac{\lambda}{\mu}$$

I02/4/152. $\int_{-1}^1 \frac{(3x^2 + 5x^4)dx}{1 + (1 + x^2 + x^4)^x}$ is equal to _____.

$\int_{-1}^1 \frac{(3x^2 + 5x^4)dx}{1 + (1 + x^2 + x^4)^x}$ का मान होगा

Ans. 4

Sol. $I = \int_{-1}^1 \frac{(3x^2 + 5x^4)dx}{1 + (1 + x^2 + x^4)^x}$

$$I = \int_{-1}^1 \frac{(3x^2 + 5x^4)[1 + x^2 + x^4]^x dx}{1 + (1 + x^2 + x^4)^x} \left(\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right)$$

$$2I = \int_{-1}^1 (3x^2 + 5x^4) dx = x^3 + x^5 \Big|_{-1}^1 = 4$$

I02/4/153. Let $f(\alpha) = \int_{\alpha^{-1}}^{\alpha} \frac{1}{x} \cot^{-1} \left(\frac{x^2 - x + 1}{2x - 3x^2} + \frac{x^2 - x + 1}{3 - 2x} \right) dx$ and

$$g(\alpha) = \int_{\ln \frac{1}{\alpha}}^{\ln \alpha} \left(\frac{|x^2 - 3x + 2| - |(x+1)(x+2)| + |x+1| + |x-1|}{|x+1| + |x-1|} \right) dx$$

where $\alpha \in (0, \infty) - \left\{ \frac{2}{3}, \frac{3}{2} \right\}$. If $f(200) - \frac{\pi}{2} g(50) = \frac{a\pi}{3} \ln b$, then find unit digit of the value of b^a .

_____.

$$\text{माना } f(\alpha) = \int_{\alpha^{-1}}^{\alpha} \frac{1}{x} \cot^{-1} \left(\frac{x^2 - x + 1}{2x - 3x^2} + \frac{x^2 - x + 1}{3 - 2x} \right) dx \text{ तथा}$$

$$g(\alpha) = \int_{\ln \frac{1}{\alpha}}^{\ln \alpha} \left(\frac{|x^2 - 3x + 2| - |(x+1)(x+2)| + |x+1| + |x-1|}{|x+1| + |x-1|} \right) dx$$

जहाँ $\alpha \in (0, \infty) - \left\{ \frac{2}{3}, \frac{3}{2} \right\}$ है। यदि $f(200) - \frac{\pi}{2} g(50) = \frac{a\pi}{3} \ln b$ है, तो b^a के मान का इकाई अंक होगा।

Ans. 64

Sol. We have $f(\alpha) = \int_{\alpha^{-1}}^{\alpha} \frac{1}{x} \cot^{-1} \left(\frac{x^2 - x + 1}{2x - 3x^2} + \frac{x^2 - x + 1}{3 - 2x} \right) dx \quad \dots(1)$

$$x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$$

$$f(\alpha) = \int_{\alpha}^{\frac{1}{\alpha}} t \cot^{-1} \left(\frac{t^2 - t + 1}{2t - 3} + \frac{t^2 - t + 1}{3t^2 - 2t} \right) \left(\frac{-1}{t^2} \right) dt = \int_{\frac{1}{\alpha}}^{\alpha} \frac{1}{t} \cot^{-1} \left(\frac{t^2 - t + 1}{2t - 3} + \frac{t^2 - t + 1}{3t^2 - 2t} \right) dt$$

$$= \int_{\frac{1}{\alpha}}^{\alpha} \frac{1}{t} \left\{ \pi - \cot^{-1} \left(\frac{t^2 - t + 1}{3 - 2t} + \frac{t^2 - t + 1}{2t - 3t^2} \right) \right\} dt \quad \dots(2)$$

$\therefore$ On equation (1) + equation (2), we get

$$2f(\alpha) = \int_{\frac{1}{\alpha}}^{\alpha} \frac{\pi}{t} = \pi \left(\ln \alpha - \ln \left(\frac{1}{\alpha} \right) \right) = 2\pi \ln \alpha$$

$$\boxed{f(\alpha) = \pi \ln \alpha}$$

Now $g(\alpha) = \int_{\ln \frac{1}{\alpha}}^{\ln \alpha} \left(\frac{|x^2 - 3x + 2| - |(x+1)(x+2)|}{|x+1| + |x-1|} + 1 \right) dx$
Odd function i.e. $f(-x) = -f(x)$

$$g(\alpha) = \int_{\ln \left(\frac{1}{\alpha} \right)}^{\ln \alpha} 1 \cdot dx = \ln \alpha - \ln \left(\frac{1}{\alpha} \right) = 2 \ln \alpha$$

$$\text{Now } f(200) - \frac{\pi}{2} g(50) = \pi \ln(200) - \pi \ln(50) = \pi \ln 4 = 3 \cdot \frac{\pi}{3} \ln 4$$

$$\text{Hence } b^a = 4^3 = 64$$

I02/4/154. Let a be a real number. If the value of definite integral $\int_{-\pi+a}^{3\pi+a} |x - a - \pi| \sin\left(\frac{x}{2}\right) dx$ is equal to -16 then sum of all integral values of a in $[0, 314]$ is $400k\pi$. Find the value of k . _____.

माना a एक वास्तविक संख्या है। यदि निश्चित समाकलन $\int_{-\pi+a}^{3\pi+a} |x - a - \pi| \sin\left(\frac{x}{2}\right) dx$ का मान -16 है तब अंतराल $[0, 314]$ में a के सभी मानों का योग $400k\pi$ है। k का मान ज्ञात कीजिए।

Ans. 1200

Sol.
$$I = \int_{-\pi+a}^{3\pi+a} |x - a - \pi| \sin\left(\frac{x}{2}\right) dx$$

Put $(x - a - \pi) = t$

$$\begin{aligned} I &= \int_{-2\pi}^{2\pi} |t| \sin\left(\frac{\pi}{2} + \frac{a+t}{2}\right) dt = \int_{-2\pi}^{2\pi} |t| \cos\left(\frac{a+t}{2}\right) dt \\ &= \underbrace{\cos \frac{a}{2} \int_{-2\pi}^{2\pi} |t| \cos\left(\frac{t}{2}\right) dt}_{\text{even}} - \underbrace{\sin \frac{a}{2} \int_{-2\pi}^{2\pi} |t| \sin\left(\frac{t}{2}\right) dt}_{\text{zero being odd}} \end{aligned}$$

$$I = 2 \cos \frac{a}{2} \int_0^{2\pi} t \cos \frac{t}{2} dt$$

Put $\frac{t}{2} = y$, we get

$$I = 8 \cos \frac{a}{2} \int_0^{\pi} \underbrace{y}_{\text{I}} \underbrace{\cos y}_{\text{II}} dy = 8 \cos \frac{a}{2} \left[\underbrace{y \sin y}_{\text{zero}} \Big|_0^{\pi} - \int_0^{\pi} \sin y dy \right]$$

$$\therefore I = -16 \cos \frac{a}{2} \Rightarrow -16 \cos \frac{a}{2} = -16$$

$$\therefore \cos \frac{a}{2} = 1 \Rightarrow \frac{a}{2} = 2n\pi \Rightarrow a = 4n\pi, n \in \mathbb{I}$$

$$\text{Sum} = \pi[4 + 8 + 12 + \dots + 96] = \pi\left[\frac{24}{2} \times 100\right] = \pi[24 \times 50] \Rightarrow k = 1200$$

I02/4/155. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the functional equation $f(x) + f\left(1 - \frac{1}{x}\right) = \tan^{-1} x$, $\forall x \in \mathbb{R} - \{0\}$ and let $N = \int_0^1 f(x) dx$, then least integer greater than or equal to N is _____.

यदि $f: \mathbb{R} \rightarrow \mathbb{R}$ में फलन समीकरण $f(x) + f\left(1 - \frac{1}{x}\right) = \tan^{-1} x$, $\forall x \in \mathbb{R} - \{0\}$ को संतुष्ट करता है तथा माना

$N = \int_0^1 f(x) dx$, तो न्यूनतम पूर्णांक जो N या N से बड़ा है, होगा –

Ans. $\frac{3\pi}{8}$

Sol. $f(x) + f\left(\frac{x-1}{x}\right) = \tan^{-1} x$ (1)

Replace x by $\frac{x-1}{x}$, we get

$$f\left(\frac{x-1}{x}\right) + f\left(\frac{1}{1-x}\right) = \tan^{-1}\left(\frac{x-1}{x}\right) \quad \text{.....(2)}$$

Replace x by $\frac{1}{1-x}$ in (1) we get

$$f\left(\frac{1}{1-x}\right) + f(x) = \tan^{-1}\left(\frac{1}{1-x}\right) \quad \text{.....(3)}$$

(1) – (2) + (3), we get

$$2f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{1-x}\right) - \tan^{-1}\left(\frac{x-1}{x}\right)$$

Also,

$$2f(1-x) = \tan^{-1}(1-x) + \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{x}{x-1}\right)$$

adding we get

$$2(f(x) + f(1-x)) = \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} \quad \forall x \in (0, 1)$$

$$= \frac{3\pi}{2}$$

$$\text{Now, } N = \int_0^1 f(x) dx = \frac{1}{2} \int_0^1 (f(x) + f(1-x)) dx = \frac{3\pi}{8}$$

I02/4/156. If $\int_0^{\frac{\pi}{2}} \frac{x \sin 2x}{\sin^4 x + \cos^4 x} dx = k \left(\int_{\pi}^{\frac{5\pi}{4}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx \right)^2$, then find the value of k. _____.

यदि $\int_0^{\frac{\pi}{2}} \frac{x \sin 2x}{\sin^4 x + \cos^4 x} dx = k \left(\int_{\pi}^{\frac{5\pi}{4}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx \right)^2$, तो k का मान होगा।

Ans. 2

Sol. Let $I_1 = \int_0^{\frac{\pi}{2}} \frac{x \sin 2x}{\sin^4 x + \cos^4 x} dx$

(Using King and Queen property)

$$\text{Also, } I_2 = k \left(\int_{\pi}^{\frac{5\pi}{4}} \left(\frac{\sin 2x}{\sin^4 x + \cos^4 x} \right) dx \right)$$

$$= k \left(\frac{\pi}{4} \right)^2 = \frac{k\pi^2}{16}$$

$$\text{So, } k = 2$$

I02/4/157. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $g(x) = \max. (\cos(\pi x), \sin(\pi x))$ for all $x \in \mathbb{R}$. The value of $\frac{\pi}{4\sqrt{2}} \int_{-10}^{10} g(x) dx$, is equal to : _____.

सभी $x \in \mathbb{R}$ के लिये माना $g: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार परिभाषित है कि $g(x) = \max. (\cos(\pi x), \sin(\pi x))$ है तो $\frac{\pi}{4\sqrt{2}} \int_{-10}^{10} g(x) dx$

का मान होगा :

Ans. 5

Sol.
$$\frac{\pi}{4\sqrt{2}} \int_{-10}^{10} g(x) dx = \frac{10\pi}{4\sqrt{2}} \int_0^2 g(x) dx$$
$$= \frac{10\pi}{4\sqrt{2}} \left[\int_0^{1/4} \cos(\pi x) dx + \int_{1/4}^{6/4} \sin(\pi x) dx + \int_{6/4}^2 \cos(\pi x) dx \right]$$
$$\frac{10\pi}{4\sqrt{2}} \left(\frac{2\sqrt{2}}{\pi} \right) = 5$$

I02/4/158. Consider the lines $L_1 : 3x^2 - 4xy - 4y^2 = 0$ and $L_2 : 15x^2 + (10 - 3k)xy - 2ky^2 = 0$. If both the lines given by L_1 coincide with lines given by L_2 , then the value of $\int_0^{2\pi} |\sin kx| dx$ is. _____.

रेखाओं $L_1 : 3x^2 - 4xy - 4y^2 = 0$ तथा $L_2 : 15x^2 + (10 - 3k)xy - 2ky^2 = 0$ पर विचार कीजिए। यदि L_1 द्वारा प्रदर्शित

रेखाएँ, L_2 द्वारा प्रदर्शित रेखाओं से मिलती हैं, तब $\int_0^{2\pi} |\sin kx| dx$ का मान होगा _____.

Ans. 4

Sol. $\therefore L_1 : 3x^2 - 4xy - 4y^2 = 0$ and
 $L_2 : 15x^2 + (10 - 3k)xy - 2ky^2 = 0$
represent same pair of lines

$$\therefore \frac{3}{15} = \frac{-4}{10 - 3K} = \frac{-4}{-2k} \Rightarrow K = 10$$

$$\therefore \int_0^{2\pi} |\sin kx| dx = \int_0^{2\pi} |\sin 10x| dx$$

use $10x = t$

$$I = \int_0^{20\pi} |\sin t| \frac{dt}{10}$$

$$I = 20 \int_0^{\pi} \sin t \frac{dt}{10}$$

$$I = 2[-\cos t]_0^{\pi}$$

$$I = 2[1 + 1] = 4$$

I02/4/159. If $\int_a^b |\sin x| dx = 8$ and $\int_0^{a+b} |\cos x| dx = 9$ then the value of $\frac{1}{\sqrt{2\pi}} \left| \int_a^b x \sin x dx \right|$ is : _____.

यदि $\int_a^b |\sin x| dx = 8$ तथा $\int_0^{a+b} |\cos x| dx = 9$ हैं तब $\frac{1}{\sqrt{2\pi}} \left| \int_a^b x \sin x dx \right|$ का मान होगा

Ans. $\frac{1}{\sqrt{2\pi}} \left| \int_{\pi/4}^{17\pi/4} x \sin x dx \right|$

Sol. $\int_a^b |\sin x| dx = 8 \Rightarrow b - a = 4\pi$

$$\int_0^{a+b} |\cos x| dx = 9 \Rightarrow a + b = \frac{9\pi}{2} \Rightarrow a = \frac{\pi}{4} \quad ; \quad b = \frac{17\pi}{4}$$

$$\frac{1}{\sqrt{2\pi}} \left| \int_{\pi/4}^{17\pi/4} x \sin x dx \right|$$

I02/4/160. The function $f(x) = \int_4^{x^2} \sqrt{9+t^2} dt$ is invertible. The value of $100(f^{-1})'(0)$ equal . _____.

फलन $f(x) = \int_4^{x^2} \sqrt{9+t^2} dt$ का प्रतिलोम संभव है, तो $100(f^{-1})'(0)$ का मान होगा –

Ans. $\frac{1}{20}$

Sol. at $y = 0, x = 2$

$$f'(x) = \sqrt{9+x^4} \cdot 2x$$

$$\therefore g'(y) = \frac{1}{f'(x)} \Big|_{x=2} = \frac{1}{2x\sqrt{9+x^4}} = \frac{1}{20}$$

I03/4/161. Let $y(x)$ be a real-valued differentiable function on the interval $(0, \infty)$ such that $y(1) = 0$ and satisfies

$$y'(x) = \ln x + 2 - \frac{y(x)}{x \ln x}. \text{ Find the value of } [y(e) - y'(e)].$$

[Note: where $[k]$ denotes largest integer less than or equal to k .] _____.

माना $y(x)$ अन्तराल $(0, \infty)$ में वास्तविक मान अवकलनीय फलन इस प्रकार है कि $y(1) = 0$ तथा समीकरण $y'(x) = \ln x +$

$2 - \frac{y(x)}{x \ln x}$ को संतुष्ट करता है, तो $[y(e) - y'(e)]$ का मान होगा

[नोट : जहाँ $[k]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है]

Ans. 0

Sol. We have $f'(x) = \ln x + 2 - \frac{f(x)}{x \ln x}$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x \ln x} = \ln x + 2 \quad \dots\dots\dots(1)$$

$$\text{Integrating factor} = e^{\int \frac{1}{x \ln x} dx} = \ln x$$

$\therefore$ Solution of the differential equation (1) is

$$y \ln x = \int \ln x (\ln x + 2) dx + C$$

$$\Rightarrow y \ln x = \int (\ln^2 x + 2 \ln x) dx + C \quad \left(\int (f(x) + x f'(x)) dx = x f(x) + C \right)$$

$$\Rightarrow y \ln x = x \ln^2 x + C$$

$$\text{As, } x = 1 \Rightarrow C = 0$$

$$\text{Hence, } y = x \ln x$$

$$f(x) = x \ln x \Rightarrow f'(x) = x \left(\frac{1}{x} \right) + \ln x$$

$$f(e) = e, f'(e) = 1 + 1 = 2$$

$$f(e) - f'(e) = e - 2 \Rightarrow [f(e) - f'(e)] = 0$$

I03/4/162. If the differential equation of a curve passing through $\left(0, -\frac{\pi}{4}\right)$ and $(t, 0)$ is

$\cos y \left(\frac{dy}{dx} + e^{-x} \right) + \sin y \left(e^{-x} - \frac{dy}{dx} \right) = e^{e^{-x}}$, then the value of $te^{e^{-t}}$ is _____.

यदि कोई वक्र जो $\left(0, -\frac{\pi}{4}\right)$ तथा $(t, 0)$ से गुजरता है उसकी अवकल समीकरण

$\cos y \left(\frac{dy}{dx} + e^{-x} \right) + \sin y \left(e^{-x} - \frac{dy}{dx} \right) = e^{e^{-x}}$ है, तो $te^{e^{-t}}$ का मान होगा

Ans. 1

Sol. $(\cos y - \sin y) dy + (\cos y + \sin y) e^{-x} dx = e^{-x} \cdot dx$

$$\cos y + \sin y = t$$

$$(\cos y - \sin y) dy = dt$$

$$\frac{dt}{dx} + t \cdot e^{-x} = e^{-x}$$

$$\text{I.F.} = e^{-e^{-x}}$$

$$\Rightarrow t \cdot e^{-e^{-x}} = x$$

I03/4/163. If solution of differential equation $\left(\left(1 + \frac{1}{x} \right) y + 2x \sin y \right) dx + (x + \ln x + x^2 \cos y) dy = 0$ is $x^a y + y \ln x + x^b \sin y + C = 0$, then $(a + b)$ is _____.

यदि अवकल समीकरण $\left(\left(1 + \frac{1}{x} \right) y + 2x \sin y \right) dx + (x + \ln x + x^2 \cos y) dy = 0$ का हल $x^a y + y \ln x + x^b \sin y +$

$C = 0$ हो, तो $(a + b)$ का मान होगा

Ans. 3

Sol. $(y dx + x dy) + \left(y \frac{dx}{x} + \ln x dy \right) + (2x \sin y dx + x^2 \cos y dy) = 0$

$$\int d(xy) + \int (y \ln x) + \int d(x^2 \sin y) = 0$$

$$\Rightarrow xy + y \ln x + x^2 \sin y + C = 0$$

$$a = 1, b = 2 \Rightarrow a + b = 3$$

I03/4/164. The solution of the differential equation $y(x^2 + y^2 + 1) dy - x(1 - 2y^2 - 2x^2) dx = 0$ is $ax^2 + y^2 + b \log(2 - x^2 - y^2) = c$ then $a + b$ is equal to _____.

अवकल समीकरण $y(x^2 + y^2 + 1) dy - x(1 - 2y^2 - 2x^2) dx = 0$ का हल $ax^2 + y^2 + b \log(2 - x^2 - y^2) = c$ है तो

$a + b$ का मान होगा

Ans. 5

Sol. Putting $x^2 + y^2 = t$

$$\Rightarrow \frac{y}{x} \cdot \frac{dy}{dx} = \frac{1}{2x} \cdot \frac{dt}{dx} - 1 - \frac{1-2t}{t+1}$$

$$\Rightarrow 2x dx = \frac{t+1}{2-t} dt$$

On integration, we get $2x^2 + y^2 + 3 \log(2 - x^2 - y^2) = c$

I01/4/165. If $P = \int e^{3x} \cos(2013x) dx$ and $Q = \int e^{3x} \sin(2013x) dx$ (where P & Q both have integration constant zero) & $f(x) = \ln(P^2 + Q^2) (3^2 + 2013^2)$ then $f(1)$ is _____.

यदि $P = \int e^{3x} \cos(2013x) dx$ तथा $Q = \int e^{3x} \sin(2013x) dx$ (जहाँ P तथा Q दोनों का समाकलन अचर शून्य है) तथा $f(x) = \ln(P^2 + Q^2) (3^2 + 2013^2)$ हो, तो $f(1)$ का मान होगा _____.

Ans. 6

Sol.
$$P = \frac{e^{3x}}{3^2 + 2013^2} (3 \cos 2013x + 2013 \sin 2013x)$$

$$Q = \frac{e^{3x}}{3^2 + 2013^2} (3 \sin 2013x - 2013 \cos 2013x)$$

$$P^2 + Q^2 = \frac{e^{6x}}{(3^2 + 2013^2)} (3^2 + 2013^2)$$

$$P^2 + Q^2 = \frac{e^{6x}}{(3^2 + 2013^2)}$$

$$\ln((P^2 + Q^2) (3^2 + 2013^2)) = 6x$$

$$f(1) = 6$$

I01/4/166. If $\int e^{x^2} (4x^2 + x^{-2}) dx = e^{x^2} (ax + bx^{-1}) + c$, where c is arbitrary constant, then find the value of a + b _____.

यदि $\int e^{x^2} (4x^2 + x^{-2}) dx = e^{x^2} (ax + bx^{-1}) + c$ है जहाँ c एक स्वेच्छ अचर है, तो a + b का मान होगा _____.

Ans. 1

Sol.
$$\int e^{x^2} (4x^2 + x^{-2}) dx$$

$$= \int e^{x^2} \cdot 2x \cdot 2x dx + \int e^{x^2} \cdot x^{-2} dx$$

$$= 2x e^{x^2} - \int 2e^{x^2} dx + \int e^{x^2} \cdot x^{-2} dx$$

$$= 2x e^{x^2} - \int x^{-1} e^{x^2} 2x dx + \int e^{x^2} \cdot x^{-2} dx = 2x e^{x^2} - x^{-1} e^{x^2} \int e^{x^2} \cdot x^{-2} dx + \int e^{x^2} \cdot x^{-2} dx$$

$$= 2xe^{x^2} - x^{-1} \cdot e^{x^2} + c = e^{x^2}(2x - x^{-1}) + c \quad \therefore a = 2, b = 1 \quad \therefore a + b = 1$$

I01/4/167. If $\int (x+1)\sqrt{1+x-x^2} dx$ is equal to $-\frac{1}{3}(1-x-x^2)^{3/2} + \frac{1}{8}(2x+1)\sqrt{1-x-x^2} + \frac{p}{16}\sin^{-1}\left(\frac{2x+1}{\sqrt{5}}\right) + C$, $p \in \mathbb{N}$ find the value of p _____.

यदि $\int (x+1)\sqrt{1+x-x^2} dx$, $-\frac{1}{3}(1-x-x^2)^{3/2} + \frac{1}{8}(2x+1)\sqrt{1-x-x^2} + \frac{p}{16}\sin^{-1}\left(\frac{2x+1}{\sqrt{5}}\right) + C$ के बराबर है, $p \in \mathbb{N}$ तो p का मान ज्ञात कीजिए _____.

Ans. 5

Sol. Let $x+1 = \lambda \cdot \frac{d}{dx}(1-x-x^2) + \mu$. Then, $x+1 = \lambda(-1-2x) + \mu$

Comparing the coefficients of like powers of x , we get $-2\lambda = 1$ and $\mu - \lambda = 1$

$$\Rightarrow \lambda = -\frac{1}{2} \text{ and } \mu = \frac{1}{2}$$

$$\therefore x+1 = -\frac{1}{2}(-1-2x) + \frac{1}{2}$$

$$\text{So, } \int (x+1)\sqrt{1-x-x^2} dx$$

$$= \int \left\{ -\frac{1}{2}(-1-2x) + \frac{1}{2} \right\} \sqrt{1-x-x^2} dx$$

$$= -\frac{1}{2} \int (-1-2x)\sqrt{1-x-x^2} dx + \frac{1}{2} \int \sqrt{1-x-x^2} dx$$

$$= \frac{1}{2} \int \sqrt{1-x-x^2} d(1-x-x^2) + \frac{1}{2} \int \sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x + \frac{1}{2}\right)^2} dx$$

$$= -\frac{1}{3}(1-x-x^2)^{3/2} + \frac{1}{8}(2x+1)\sqrt{1-x-x^2} + \frac{5}{16}\sin^{-1}\left(\frac{2x+1}{\sqrt{5}}\right) + C$$

MATCH THE COLUMN TYPE QUESTION(S)

I02/5/168. Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) If $f(x) + f\left(\frac{1}{x}\right) = F(x)$, where $f(x) = \int_1^x \frac{\ln t}{1+t} dt$,

(p) 3

then $F(e^2)$ is equal to

(b) If $f(x)$ is differentiable and $\int_0^{x^2} t f(t) dt = \frac{2}{5} x^5$ then $f(9)$ (q) 7

is equal to

(c) Let $F(x) = \int_0^{x^2} f(t) dt$. If $F(x) = x^2 + x^3$, then $f(16)$ is equal to (r) 2

(d) If $\int_{\cos x}^1 t^2 f(t) dt = 1 - \cos x$, then $f\left(\frac{1}{\sqrt{5}}\right)$ is equal to (s) 5

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) यदि $f(x) + f\left(\frac{1}{x}\right) = F(x)$, जहाँ $f(x) = \int_1^x \frac{\ln t}{1+t} dt$ है, तो (p) 3

$F(e^2)$ का मान होगा

(b) यदि $f(x)$ अवकलनीय तथा $\int_0^{x^2} t f(t) dt = \frac{2}{5} x^5$ हो, तो $f(9)$ (q) 7

का मान होगा

(c) माना $F(x) = \int_0^{x^2} f(t) dt$ है। यदि $F(x) = x^2 + x^3$ हो, तो $f(16)$ का मान होगा (r) 2

(d) यदि $\int_{\cos x}^1 t^2 f(t) dt = 1 - \cos x$ हो, तो $f\left(\frac{1}{\sqrt{5}}\right)$ का मान होगा (s) 5

सही विकल्प का चयन कीजिए :

(A) a-p, b-q, c-r, d-s (B) a-r, b-p, c-q, d-s (C) a-s, b-r, c-q, d-p (D) a-p, b-r, c-s, d-q

Ans. B

Sol. P. $f(x) = \int_1^x \frac{\ln t}{1+t} dt$

$$f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\ln t}{1+t} dt$$

Put $t = \frac{1}{z}$

$$f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln(z)}{z(1+z)} dz$$

$$F(x) = f(x) + f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln z}{1+z} \left(1 + \frac{1}{z}\right) dz$$

$$= \int_1^x \frac{\ln z}{z} dz = \frac{(\ln x)^2}{2}$$

$$F(e^2) = \frac{4}{2} = 2$$

Q. Differentiate Both side wrt.x

$$x^2 f(x^2).2x = 2x^4$$

$$f(x^2) = x$$

$$\text{Put } x = 3$$

$$f(9) = 3$$

R. Differentiate w.r.t x

$$F'(x) = f(x^2).2x = 2x + 3x^2$$

$$f(x^2) = \frac{2+3x}{2}$$

$$x = 4$$

$$f(16) = \frac{2+12}{2} = 7$$

S. Differentiate w.r.t. x

$$-\cos^2 x f(\cos x)(-\sin x) = \sin x$$

$$\Rightarrow f(\cos x) = \frac{1}{\cos^2 x}$$

$$\cos x = \frac{1}{\sqrt{5}}$$

$$f\left(\frac{1}{\sqrt{5}}\right) = 5$$

102/5/169. Match the following column :

Column-I

$$(a) \int_1^5 \sqrt{x-2} \sqrt{x-1} dx$$

$$(b) \int_{-1}^2 \left[\frac{[x]}{1+x^2} \right] dx \text{ where } [.] \text{ denotes greatest integer function}$$

Column-II

$$(p) \frac{\pi}{2}$$

$$(q) 2$$

$$(c) \int_{1/2}^2 \frac{x \ln x}{(1+x^2)^2} dx \quad (r) -1$$

$$(d) \int_0^\pi \left| \sin x - \frac{2x}{\pi} \right| dx \quad (s) 0$$

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

$$(a) \int_1^5 \sqrt{x-2\sqrt{x-1}} dx \quad (p) \frac{\pi}{2}$$

$$(b) \int_{-1}^2 \left[\frac{[x]}{1+x^2} \right] dx \text{ जहाँ } [.] \text{ महत्तम पूर्णांक फलन दर्शाता है} \quad (q) 2$$

$$(c) \int_{1/2}^2 \frac{x \ln x}{(1+x^2)^2} dx \quad (r) -1$$

$$(d) \int_0^\pi \left| \sin x - \frac{2x}{\pi} \right| dx \quad (s) 0$$

सही विकल्प का चयन कीजिए :

(A) a-q, b-r, c-s, d-p (B) a-p, b-r, c-s, d-p (C) a-p, b-q, c-r, d-s (D) a-r, b-s, c-p, d-q

Ans. A

Sol. P. $\int_1^5 \sqrt{x-2\sqrt{x-1}} dx = \int_1^5 \left| \sqrt{x-1} - 1 \right| dx$

$$= \int_1^2 (1 - \sqrt{x-1}) dx + \int_2^5 (\sqrt{x-1} - 1) dx$$

$$= \left[x - \frac{2(x-1)^{3/2}}{3} \right]_1^2 + \left[\frac{2(x-1)^{3/2}}{3} - x \right]_2^5$$

$$= \left(2 - \frac{2}{3} - 1 + 0 \right) + \left(\frac{16}{3} - 5 - \frac{2}{3} + 2 \right)$$

$$= -2 + \frac{12}{3} = 2$$

$$Q. \int_{-1}^2 \left[\frac{[x]}{1+x^2} \right] dx = \int_{-1}^0 \left[\frac{-1}{1+x^2} \right] dx + \int_0^1 [0] dx + \int_1^2 \left[\frac{1}{1+x^2} \right] dx$$

$$= \int_{-1}^0 (-1) dx + 0 + 0 \Rightarrow -1$$

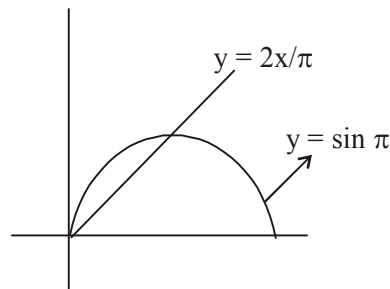
$$R. \quad I = \int_{1/2}^2 \frac{x \ln x}{(1+x^2)^2} dx \quad \text{Put } x = \frac{1}{t}$$

$$\Rightarrow dx = -\frac{1}{t^2} dt$$

$$I = - \int_2^{1/2} \frac{1}{t} \frac{\ln t}{(t^2+1)^2} \left(-\frac{1}{t^2}\right) dt$$

$$\Rightarrow I = -I$$

$$\Rightarrow I = 0$$



$$S. \quad \int_0^\pi \left| \sin x - \frac{2x}{\pi} \right| dx$$

$$\Rightarrow \int_0^{\pi/2} \left(\sin x - \frac{2x}{\pi} \right) dx + \int_{\pi/2}^\pi \left(\frac{2x}{\pi} - \sin x \right) dx$$

$$= \left| -\cos x - \frac{x^2}{\pi} \right|_0^{\pi/2} + \left| \frac{x^2}{\pi} + \cos x \right|_{\pi/2}^\pi = \frac{\pi}{2}$$

I02/5/170. Match the following column :

Column-I

(a) Number of solution(s) of the equation $[x] = |2x - 1|$ is
[where $[.]$ is greatest integer function]

(b) If $\int_0^{2\pi} \frac{1}{1 + \tan^4 x} dx = \frac{\pi}{k}$ ($k \in \mathbb{N}$), then k is equal to

(c) Value of $\int_0^\infty \frac{\ln x}{1+x^2} dx$ is equal to

(d) If $y = \sin^{-1} \cos(\sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$

then $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ is equal to

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

(a) समीकरण $[x] = |2x - 1|$ के हलों की संख्या है

Column-II

(p) 1

(q) 0

(r) 3

(s) 2

स्तम्भ-II

(p) 1

[जहाँ [.] महत्तम पूर्णांक फलन है]

(b) यदि $\int_0^{2\pi} \frac{1}{1 + \tan^4 x} dx = \frac{\pi}{k} (k \in \mathbb{N})$, तब k बराबर है (q) 0

(c) $\int_0^\infty \frac{\ln x}{1+x^2} dx$ का मान बराबर होगा (r) 3

(d) यदि $y = \sin^{-1} \cos(\sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$ (s) 2

तब $x = \frac{\pi}{6}$ पर $\frac{dy}{dx}$ होगा

सही विकल्प का चयन कीजिए :

(A) $a - q; b - s; c - p; d - r$

(B) $a - r; b - p; c - q; d - s$

(C) $a - s; b - p; c - q; d - q$

(D) $a - s; b - q; c - r; d - p$

Ans.

C

Sol.

(P) $LHS \in \text{Integer}$

$\Rightarrow x = \frac{1}{2}$ & $x = 1$ is answer.

(Q) $I = 2 \int_0^\pi \frac{dx}{1 + \tan^4 x} = 4 \int_0^{\pi/2} \frac{dx}{1 + \tan^4 x}$

Use king property

$2I = 4 \int_0^{\pi/2} dx \Rightarrow I = \pi$

$\therefore k = 1$

(R) Put $x = \frac{1}{t}$ & add two integral,

we get answer = 0

(S) $y = \sin^{-1} \cos(\cos^{-1} \sqrt{1-x^2})$

$+ \cos^{-1} \sin(\sin^{-1} \sqrt{1-x^2}) = \frac{\pi}{2}$

$\therefore \frac{dy}{dx} = 0$

I02/5/171. Match the following column :

Column-I

Column-II

(a) Value of $\int_0^1 \frac{dx}{\sqrt{1-x^2} (1+\sqrt{1-x^2})}$ is equal to (p) 2

(b) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^5 + e^{\frac{x}{2}} + e^{x^3}$ & $g(x) = f^{-1}(x)$, then value of $g'(2)$ is equal to (q) 4

(c) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{2\sqrt{n}} \right)$ equal to (r) 12

(d) If $f(x) = \int_{x-1}^{2x} (t^3 - 2t + 4) dt$, then value of $f'(1)$ is equal to (s) 1

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) $\int_0^1 \frac{dx}{\sqrt{1-x^2} (1+\sqrt{1-x^2})}$ का मान बराबर होगा (p) 2

(b) $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार परिभाषित है कि $f(x) = x^5 + e^{\frac{x}{2}} + e^{x^3}$ तथा $g(x) = f^{-1}(x)$ है, तब $g'(2)$ का मान बराबर होगा (q) 4

(c) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{2\sqrt{n}} \right)$ बराबर होगा (r) 12

(d) यदि $f(x) = \int_{x-1}^{2x} (t^3 - 2t + 4) dt$, तब $f'(1)$ का मान बराबर होगा (s) 1

सही विकल्प का चयन कीजिए :

(A) $a - s; b - p; c - q; d - r$

(B) $a - p; b - q; c - s; d - r$

(C) $a - q; b - p; c - r; d - s$

(D) $p - r; b - s; c - q; d - p$

Ans. A

Sol. (P) Put $x = \sin \theta$

$$\therefore \int_0^{\pi/2} \frac{d\theta}{1 + \cos \theta} = 1$$

$$(Q) f'(x) = 5x^4 + \frac{1}{2}e^{x/2} + 3x^2e^{x^3} > 0$$

$\therefore f$ is one-one map

$$g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

$$\& f(0) = 2$$

$$\therefore g'(2) = 2$$

$$(R) T_r = \frac{1}{\sqrt{n}\sqrt{r}} = \frac{1}{n(\sqrt{r/n})}$$

$$\therefore \text{Sum} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \frac{1}{\sqrt{r/n}} = \int_0^4 \frac{1}{\sqrt{x}} dx = 4$$

$$(S) f'(x) = [(2x)^3 - 2(2x) + 4] \times 2$$

$$-[(x-1)^3 - 2(x-1) + 4]$$

$$f'(1) = 12$$

I01/5/172. If $y^2 = 3x^2 + 2x + 1$ & integration I_n is defined as $I_n = \int \frac{x^n}{y} dx$ where $AI_{10} + BI_9 + CI_8 = x^9 y$ then :

Column-I

- (a) A is equal to
- (b) B is equal to
- (c) C is equal to
- (d) Absolute value of thrice of x intercept of the line (s)
 $Cx + By + A = 0$

Column-II

- (p) 9
- (q) 10
- (r) 19
- (s) 30

Choose the correct option :

माना $y^2 = 3x^2 + 2x + 1$ तथा I_n , $I_n = \int \frac{x^n}{y} dx$ जहाँ $AI_{10} + BI_9 + CI_8 = x^9 y$ है।

निम्न कॉलम का सुमेलित कीजिए :

स्तम्भ-I

- (a) A का मान होगा
- (b) B का मान होगा
- (c) C का मान होगा
- (d) रेखा $Cx + By + A = 0$ के x निर्देशांक के तीन गुना का
निरपेक्ष मान होगा

स्तम्भ-II

- (p) 9
- (q) 10
- (r) 19
- (s) 30

सही विकल्प का चयन कीजिए :

- (A) a-p, b-s, c-r, d-q
- (C) a-s, b-r, c-p, d-q

- (B) a-q, b-r, c-s, d-p
- (D) a-p, b-q, c-r, d-s

Ans.

C

Sol.

$$y^2 = 3x^2 + 2x + 1$$

$$I_n = \int \frac{x^n}{y} dx$$

$$AI_{10} + BI_9 + CI_8 = x^9 y$$

$$A \int \frac{x^{10}}{y} dx + B \int \frac{x^9}{y} dx + C \int \frac{x^8}{y} dx = x^9 \cdot y$$

differentiate both sides.

$$\frac{Ax^{10}}{y} + \frac{Bx^9}{y} + \frac{Cx^8}{y} = x^9 \frac{dy}{dx} + y \cdot 9x^8$$

$$Ax^{10} + Bx^9 + Cx^8 = x^9 \cdot y \frac{dy}{dx} + 9y^2 x^8$$

$$Ax^{10} + Bx^9 + Cx^8 = x^9(3x + 1) + 9x^8(3x^2 + 2x + 1)$$

Compare the coefficients

$$\Rightarrow A = 30$$

$$B = 19$$

$$C = 9$$

$$\Rightarrow \text{x-int of } Cx + By + A = 0 \Rightarrow -\frac{A}{C}$$

$$\left| \frac{3A}{C} \right| = 10$$

I01/5/173.

Column-I

Column-II

(a) If $\int \frac{dx}{x^{1/3} + 2} = g(x) + C$ where $g(0) = 12 \ln 2$,

(p) 1

then $[g(-1)]$ is equal to

(where $[.]$ denotes greatest integer function)

(b) If $\int (\sec x)^{9/5} (\operatorname{cosec} x)^{1/5} dx = k \tan^m x + C$, (q)

3

then km is equal to

(c) Let $\int \frac{dx}{\cot^2 x - 1} = \frac{1}{\ell} \ln |\sec 2x + \tan 2x| - \frac{x}{m} + C$,

(r) 6

then $\ell + m$ is equal to

(d) Let $\int \left(1 + \frac{1}{x}\right) dx = \ell \ln \left| \frac{g(x)-1}{g(x)+1} \right| + C$, then $[g^2(1)]$ is equal to

(s) 7

(where $[.]$ denotes greatest integer function)

Choose the correct option :

स्तम्भ-I

स्तम्भ-II

(a) यदि $\int \frac{dx}{x^{1/3} + 2} = g(x) + C$ जहाँ $g(0) = 12 \ln 2$ हो, तो $g(-1)$ का मान होगा (p) 1

(b) यदि $\int (\sec x)^{9/5} (\cos \sec x)^{1/5} dx = k \tan^m x + C$ हो, तो km का मान होगा (q) 3

(c) माना $\int \frac{dx}{\cot^2 x - 1} = \frac{1}{\ell} \ln |\sec 2x + \tan 2x| - \frac{x}{m} + C$ हो, (r) 6

तो $l + m$ का मान होगा

(d) माना $\int \frac{\left(1 + \frac{1}{x}\right) dx}{\sqrt{1 + xe^x}} = \ell \ln \left| \frac{g(x) - 1}{g(x) + 1} \right| + C$ हो, तो $[g^2(1)]$ का मान होगा (s) 7

जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है।

सही विकल्प का चयन कीजिए :

(A) a-s, b-r, c-q, d-p

(B) a-s, b-p, c-q, d-r

(C) a-p, b-q, c-r, d-s

(D) a-s, b-p, c-r, d-q

Ans. D

Sol. (A) $\int \frac{dx}{x^{1/3} + 2}$

Let $x = t^3$

$dx = 3t^2 dt$

$I = 3 \int \frac{t^2 dt}{t + 2}$

$\Rightarrow \frac{I}{3} = \int \frac{(t^2 - 4 + 4) dt}{t + 2}$

$\Rightarrow \frac{I}{3} = \int \left((t^2 - 2) + \frac{4}{t + 2} \right) dt$

$\Rightarrow \frac{I}{3} = \frac{t^3}{3} - 2t + 4 \ln |t + 2|$

$\Rightarrow I = \frac{3}{2} x^{2/3} - 6x^{1/3} + 12 \ln |x^{1/3} + 2| + C$

$g(x) = \frac{3}{2} x^{2/3} - 6x^{1/3} + 12 \ln |x^{1/3} + 2|$

$g(-1) = \frac{3}{2} + 6 = \frac{15}{2} \quad [g(-1)] = 7$

$$(B) \quad I = \int (\sec x)^{9/5} (\operatorname{cosec} x)^{1/5} dx$$

$$= \int \frac{\sec^2 x \, dx}{(\tan x)^{1/5}}$$

$$\text{Let } \tan x = t^5$$

$$\sec^2 x \, dx = 5 t^4 dt$$

$$\Rightarrow I = 5 \int \frac{t^4 dt}{t} = \frac{5 t^4}{4} = \frac{5}{4} (\tan x)^{4/5} + C \Rightarrow km = 1$$

$$(C) \quad \int \frac{dx}{\cot^2 x - 1} = \int \frac{\sin^2 x dx}{\cos 2x}$$

$$2I = \int \frac{(1 - \cos 2x) dx}{\cos 2x}$$

$$= \int \sec 2x dx - \int dx$$

$$2I = \frac{1}{2} \ell n |\sec 2x + \tan 2x| - x$$

$$\Rightarrow I = \frac{1}{4} \ell n |\sec 2x + \tan 2x| - \frac{x}{2} + C$$

$$\Rightarrow \ell + m = 6$$

$$(D) \quad I = \int \frac{(x+1)e^x dx}{xe^x \sqrt{1+xe^x}}$$

$$\text{Let } 1 + xe^x = t^2$$

$$\Rightarrow (x+1)e^x dx = 2t dt$$

$$\Rightarrow I = 2 \int \frac{t dt}{(t^2-1)t} = \ell n \left| \frac{t-1}{t+1} \right| + C = \ell n \left| \frac{\sqrt{1+xe^x}-1}{\sqrt{1+xe^x}+1} \right| + C$$

$$\Rightarrow g^2(x) = 1 + xe^x \Rightarrow g^2(1) = 1 + e$$

$$\Rightarrow g^2(1) = 3.71828 \Rightarrow [g^2(1)] = 3$$

I01/5/174.

Column-I

Column-II

$$(a) \text{ If } \int \frac{dx}{x^{1/3} + 2} = g(x) + C \text{ where } g(0) = 12 \ln 2, \text{ then } g(-1) \text{ is equal to } (p) \ 1$$

$$(b) \text{ If } \int (\sec x)^{9/5} (\operatorname{cosec} x)^{1/5} dx = k \tan^m x + C, \text{ then } km \text{ is equal to } (q) \ 3$$

$$(c) \text{ Let } \int \frac{dx}{\cot^2 x - 1} = \frac{1}{\ell} \ell n |\sec 2x + \tan 2x| - \frac{x}{m} + C, (r) 6$$

then $\ell + m$ is equal to

(d) Let $\int \frac{\left(1 + \frac{1}{x}\right) dx}{\sqrt{1 + xe^x}} = \ell n \left| \frac{g(x)-1}{g(x)+1} \right| + C$, then $[g^2(1)]$ is equal to

(s) Rational number

(t) Irrational number

Where $[.]$ denote greatest integer function

Column-I

Column-II

(a) यदि $\int \frac{dx}{x^{1/3} + 2} = g(x) + C$ जहाँ $g(0) = 12/\ln 2$ हो, तो $g(-1)$ का मान होगा

(p) 1

(b) यदि $\int (\sec x)^{9/5} (\cos \sec x)^{1/5} dx = k \tan^m x + C$ हो, तो km का मान होगा

(q) 3

(c) माना $\int \frac{dx}{\cot^2 x - 1} = \frac{1}{\ell} \ell n |\sec 2x + \tan 2x| - \frac{x}{m} + C$ हो,

(r) 6

तो $l + m$ का मान होगा

(d) माना $\int \frac{\left(1 + \frac{1}{x}\right) dx}{\sqrt{1 + xe^x}} = \ell n \left| \frac{g(x)-1}{g(x)+1} \right| + C$ हो, तो $[g^2(1)]$ का मान होगा

(s) परिमेय संख्या

(t) अपरिमेय संख्या

जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है।

(A) a-s; b-p,s; c-r,s; d-q,s

(B) a-t; b-p, t; c-r,s; d-q,s

(C) a-s; b-p,s; c-r, t; d-p, s

(D) a-s; b-p,s; c-p,q; d-p,s

Ans. A

Sol. (A) $\int \frac{dx}{x^{1/3} + 2}$

Let $x = t^3$

$dx = 3t^2 dt$

$I = 3 \int \frac{t^2 dt}{t+2}$

$\Rightarrow \frac{I}{3} = \int \frac{(t^2 - 4 + 4) dt}{t+2}$

$\Rightarrow \frac{I}{3} = \int \left((t^2 - 2) + \frac{4}{t+2} \right) dt$

$$\Rightarrow \frac{I}{3} = \frac{t^2}{2} - 2t + 4 \ln |t + 2|$$

$$\Rightarrow \frac{I}{3} = \frac{3}{2} x^{2/3} - 6x^{1/3} + 12 \ln |x^{1/3} + 2| + C$$

$$g(x) = \frac{3}{2} x^{2/3} - 6x^{1/3} + 12 \ln |x^{1/3} + 2|$$

$$g(-1) = \frac{3}{2} + 6 = \frac{15}{2}$$

$$(B) I = \int (\sec x)^{9/5} (\operatorname{cosec} x)^{1/5} dx$$

$$= \int \frac{\sec^2 x dx}{(\tan x)^{1/5}}$$

$$\text{Let } \tan x = t^5$$

$$\sec^2 x dx = 5t^4 dt$$

$$\Rightarrow I = 5 \int \frac{t^4 dt}{t} = \frac{5t^4}{4} = \frac{5}{4} (\tan x)^{4/5} + C$$

$$\Rightarrow km = 1$$

$$(C) \int \frac{dx}{\cot^2 x - 1} = \int \frac{\sin^2 x dx}{\cos 2x}$$

$$2I = \int \frac{(1 - \cos 2x) dx}{\cos 2x}$$

$$= \int \sec 2x dx - \int dx$$

$$2I = \frac{1}{2} \ln |\sec 2x + \tan 2x| - x$$

$$\Rightarrow I = \frac{1}{4} \ln |\sec 2x + \tan 2x| - \frac{x}{2} + C$$

$$\Rightarrow \ell + m = 6$$

$$(D) I = \int \frac{(x+1)e^x dx}{x^x \sqrt{1+x^x}}$$

$$\text{Let } 1 + xe^x = t^2$$

$$\Rightarrow (x+1)e^x dx = 2t dt$$

$$\Rightarrow I = 2 \int \frac{t dt}{(t^2 - 1)t}$$

$$= \ln \left| \frac{t-1}{t+1} \right| + C = \ln \left| \frac{\sqrt{1+xe^x} - 1}{\sqrt{1+xe^x} + 1} \right| + C$$

$$\Rightarrow g^2(x) = 1 + xe^x \Rightarrow g^2(1) = 1 + e$$

$$\Rightarrow g^2(1) = 3.71828 \Rightarrow [g^2(1)] = 3$$

I01/5/175.

Column-I

Column-II

(a) If $\int \frac{(x^2 + 4)dx}{x^3 + 4x^2 + 4x} = a \ln |x| + \frac{b}{x+2} + C$, then $b - a$ is more than (p) 1

(b) If $f(x) = x \ln x$, then $2(f^{-1})'(\ln 4)$ is more than (q) 2
(where $f'(x)$ denotes derivative of $f(x)$ w.r.t. x)

(c) $\lim_{x \rightarrow \infty} (x \ln x)^{\frac{1}{x^2+1}}$ is equal to (r) 3

(d) If $f(x) = ||x - 1| - 1|$, then number of points where $f(x)$ is NOT differentiable is (s) 0

(t) 5

Column-I

Column-II

(a) यदि $\int \frac{(x^2 + 4)dx}{x^3 + 4x^2 + 4x} = a \ln |x| + \frac{b}{x+2} + C$ हो, तो $b - a$ का मान (p) 1

से अधिक होगा

(b) यदि $f(x) = x \ln x$ हो, तो $2(f^{-1})'(\ln 4)$ का मान से अधिक होगा (q) 2
(जहाँ $f'(x)$, x के सापेक्ष $f(x)$ के अवकलज को निरूपित करता है)

(c) $\lim_{x \rightarrow \infty} (x \ln x)^{\frac{1}{x^2+1}}$ का मान होगा (r) 3

(d) यदि $f(x) = ||x - 1| - 1|$ हो, तो बिन्दुओं की संख्या जहाँ $f(x)$ अवकलनीय नहीं हो, होगी (s) 0

(t) 5

(A) $a - rt, b - qs, c - p, d - t$

(B) $a - pqs, b - ps, c - p, d - t$

(C) $a - qrt, b - rt, c - p, d - t$

(D) $a - pt, b - rs, c - q, d - t$

Ans.

B

Sol.

(A) $\int \frac{x^2 + 4}{x(x+2)^2} dx = \int \frac{(x+2)^2 - 4x}{x(x+2)^2} dx$

$$\int \frac{1}{x} dx - 4 \int \frac{1}{(x+2)^2} dx$$

$$= \ln x + \frac{4}{x+2} + C \quad \Rightarrow \quad a=1, b=4$$

$$(B) f(x) = x \ln x = y \text{ (let)} \Rightarrow \frac{dy}{dx} = 1 + \ln x$$

$$\Rightarrow f^{-1}(y) = x \Rightarrow (f^{-1})'(y) \frac{dy}{dx} = 1$$

$$\Rightarrow (f^{-1})'(y) = \frac{1}{1 + \ln x}$$

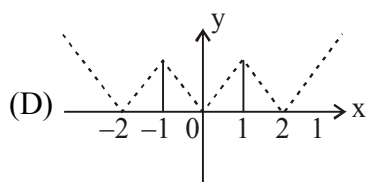
$$\because y = \ln 4 \Rightarrow x = 2$$

$$\Rightarrow (f^{-1})(\ln 4) = \frac{1}{1 + \ln 2}$$

$$\Rightarrow 2(f^{-1})(\ln 4) = \frac{2}{1 + \ln 2} > 1$$

$$(C) \lim_{x \rightarrow \infty} (x \ln x)^{\frac{1}{x^2+1}} (\infty^0) = e^{\lim_{x \rightarrow \infty} \frac{\ln(x \ln x)}{x^2+1}} \left(\frac{\infty}{\infty} \right)$$

$$= e^{\lim_{x \rightarrow \infty} \frac{1 + \ln x}{(x \ln x) 2x} \left(\frac{\infty}{\infty} \right)} = e^{\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{4x \ln 4 + 2x}} = e^0 = 1$$



$f(x) = ||x|-1|-1|$ is not differentiable at
 $x = -2, -1, 0, 1, 2$

I01/5/176.

Column-I

Column-II

(a) If the angle between the tangents from (1, 3) to the

(p) 0

curve $y^2 = 4a(x-1)$ is $\frac{5\pi}{6}$, then $a^{2/3}$ is

(b) $\left[\lim_{x \rightarrow 0} \frac{\log(e^x + 3x - 5x^2 + 9x^3)}{\log(\cos x - x + 3x^2 - 6x^3)} \right]$ is

(q) 1

(where $[.]$ denotes greatest integer function)

(c) If $h(x) = \int \frac{(1 - xe^x)dx}{xe^x + e^x + x + 1}$ & $h(0) = \ln\left(\frac{1}{2}\right)$, (r) 2

then $\lim_{x \rightarrow \infty} e^{h(x)}$ is

(d) Sum of intercepts made by any tangent of (s) 3

$$y = (\sqrt{x} + 2)^2 - 2\sqrt{x}(2 + \sqrt{x} + \sqrt{y})$$

on coordinate axes is

(t) 4

स्तम्भ-I

स्तम्भ-II

(a) यदि वक्र $y^2 = 4a(x - 1)$ पर बिन्दु $(1, 3)$ से खींची गई स्पर्श रेखाओं के (p) 0

बीच का कोण $\frac{5\pi}{6}$ है, तो $a^{2/3}$ होगा

(b) $\left[\lim_{x \rightarrow 0} \frac{\log(e^x + 3x - 5x^2 + 9x^3)}{\log(\cos x - x + 3x^2 - 6x^3)} \right]$ है (q) 1

(जहाँ $[.]$ महतम पूर्णांक फलन को दर्शाता है)

(c) यदि $h(x) = \int \frac{(1 - xe^x)dx}{xe^x + e^x + x + 1}$ तथा $h(0) = \ln\left(\frac{1}{2}\right)$, (r) 2

तो $\lim_{x \rightarrow \infty} e^{h(x)}$ का मान है

(d) $y = (\sqrt{x} + 2)^2 - 2\sqrt{x}(2 + \sqrt{x} + \sqrt{y})$ के द्वारा निदर्शी अक्षों पर (s) 3

काटे गये अन्तःखण्डों का योग है

(t) 4

(A) a-t, b-r, c-q, d-s

(B) a-p, b-q, c-r, d-s

(C) a-s, b-t, c-p, d-t

(D) a-q, b-p, c-t, d-r

Ans. C

Sol. (A) One of the tangents is tangent at vertex.

$\Rightarrow$ slope of the other tangent is

$$\tan\left(\frac{5\pi}{6} - \frac{\pi}{2}\right) = \sqrt{3}$$

$$y = m(x - 1) + \frac{a}{m} \Rightarrow 3 = \frac{a}{\sqrt{3}} \Rightarrow a = 3^{3/2} \Rightarrow a^{2/3} = 3$$

(B) $\lim_{x \rightarrow 0} \frac{\log(1 + e^x - 1 - 3x - 5x^2 + 9x^3)}{\log(1 + \cos x - 1 - x + 3x^2 - 6x^3)}$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1 - 3x - 5x^2 + 9x^3}{\cos x - 1 - x + 3x^2 - 6x^3} = -4$$

$$(C) \quad h(x) = \int \left(\frac{1}{1+x} - \frac{e^x}{1+e^x} \right) dx$$

$$h(x) = \ln \left(\frac{1+x}{1+e^x} \right) + C$$

$$h(0) = \ln \left(\frac{1}{2} \right) \Rightarrow C = 0$$

$$\text{So } h(x) = \ln \left(\frac{1+x}{1+e^x} \right)$$

$$\text{Now } \lim_{x \rightarrow \infty} \frac{1+x}{1+e^x} = 0$$

$$y = x + 4 + 4\sqrt{x} - 4\sqrt{x} - 2x - 2\sqrt{x}\sqrt{y}$$

$$\Rightarrow x + y + 2\sqrt{x}\sqrt{y} = 4 \Rightarrow \sqrt{x} + \sqrt{y} = 2$$

Tangent at any point (x_1, y_1) is

$$y - y_1 = -\frac{\sqrt{y_1}}{\sqrt{x_1}}(x - x_1)$$

$$\text{Sum of intercepts} = x_1 + y_1 + 2\sqrt{x_1}\sqrt{y_1}$$